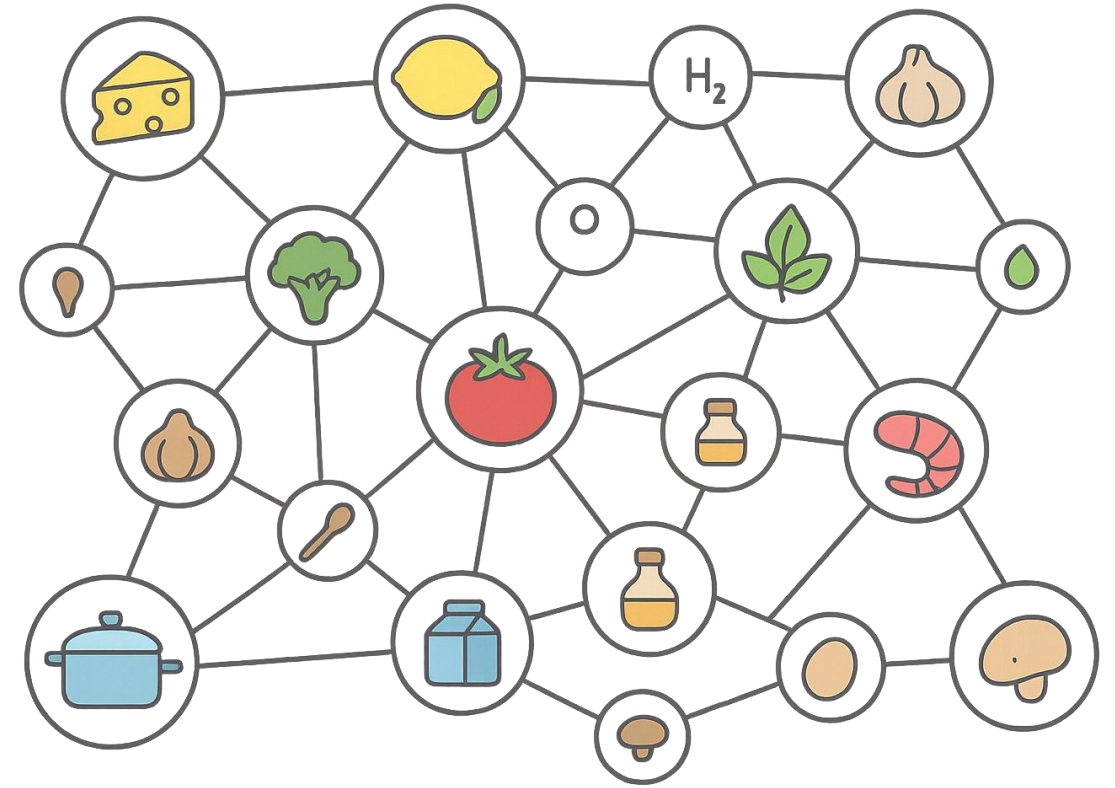


Computational Gastronomy and the Evolution of Food



Gastronomia



Arcestrato, poeta greco del 4° sec. a. C



Ingredienti Ricette

Pesci pregiati Tecniche



Siracusa Efeso

Chio Bisanzio

Conoscere il cibo



Conoscere il mondo

Gastronomia



Conoscere il cibo $\longrightarrow$ Conoscere il mondo

gastronomìa gr. γαστρονομία

(γαστήρ «stomaco» e -νομία «-legge»)



Studia la interazione tra uomo e cibo

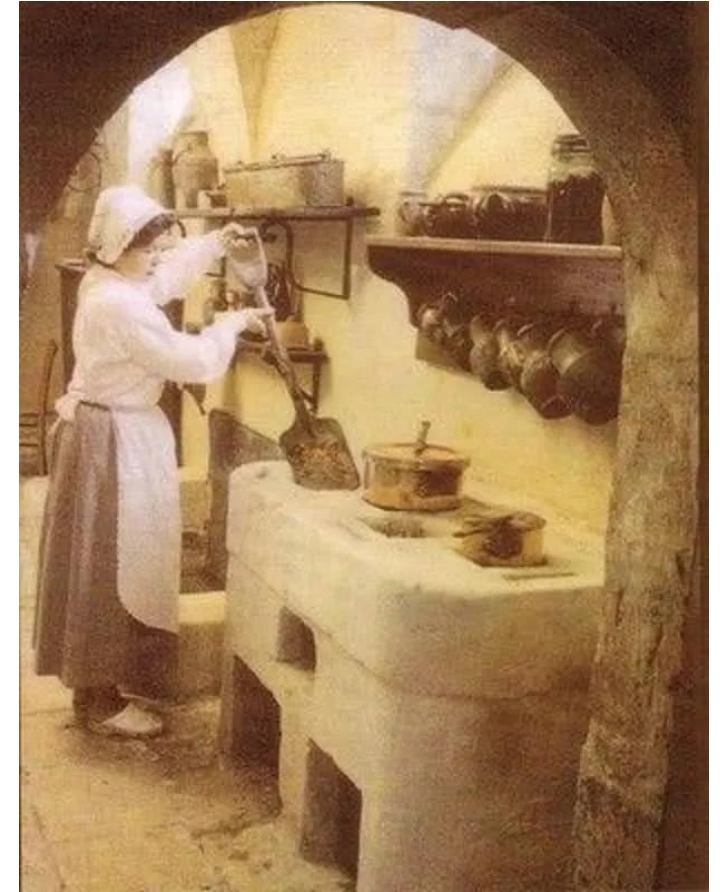
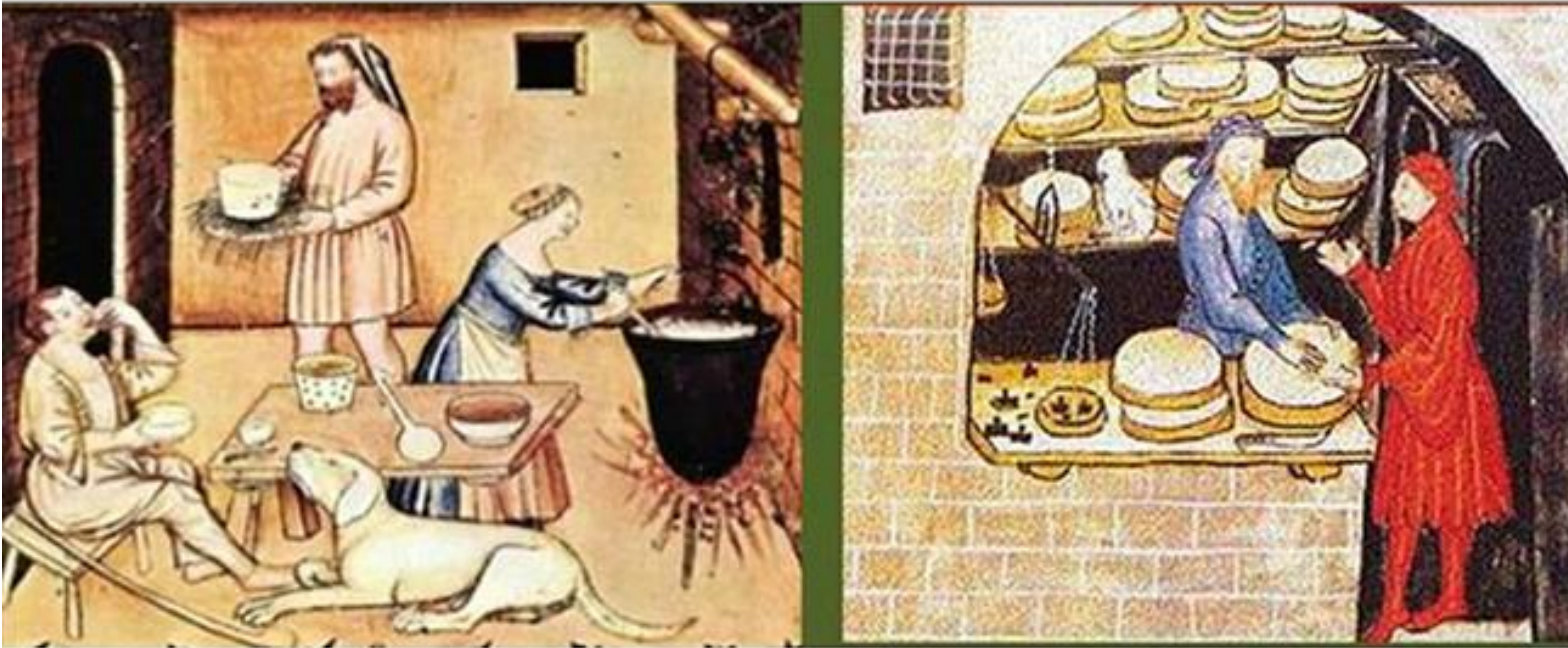
$\neq$

Scienza degli alimenti

Cosa è cucinare?

Cucinare è una tecnica (τέχνη)

Cucinare è il **processo di trasformare** ingredienti grezzi in pietanze



Cucinare rende umani

LUDWIG FEUERBACH
**L'UOMO È CIÒ
CHE MANGIA**

Saggio introduttivo e cura di Andrea Tagliapietra



Bollati Boringhieri

“L'uomo è ciò che mangia”

Unità dell'essere umano sta proprio
nell'alimentazione

Unione tra **natura** e **cultura**

Natura ↔ **Cultura**

- Sopravvivenza
- Ingredienti Grezzi
- Abitudini
- Usi e Costumi
- Geografia



Ludwig Feuerbach (1804 – 1872)

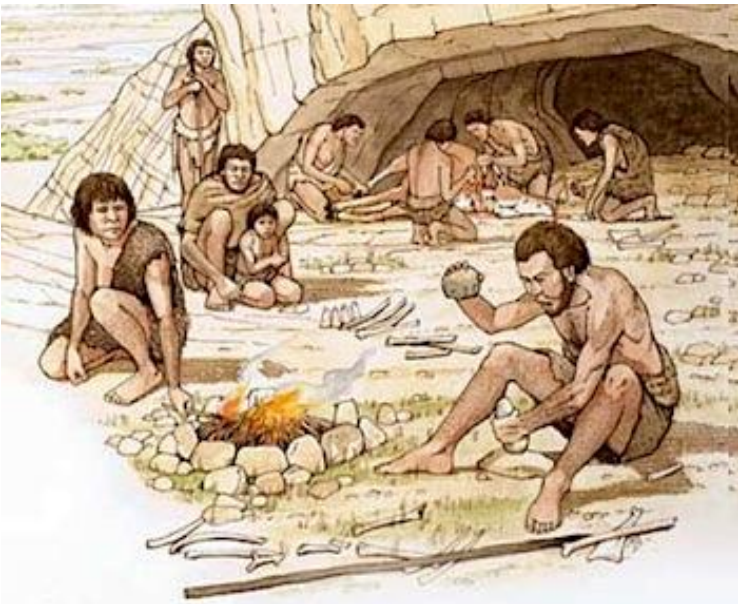
bios

significava vita concreta dell'individuo,
ma anche generi di sopravvivenza

Cucinare rende umani

“Il crudo e il cotto” (1964)

Sostiene che il simbolo del passaggio dalla natura alla cultura



Passaggio dal crudo al cotto



Trasformazione culturale

Codice simbolico che
trasmette strutture mentali



Trasformare il cibo genera cultura

COTTURA

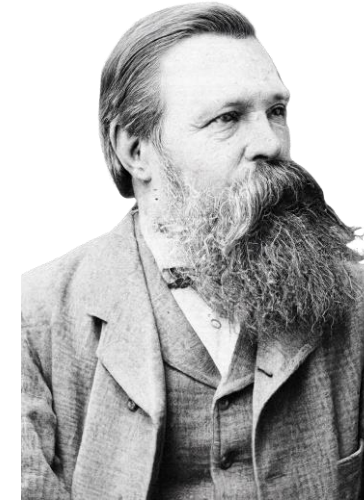
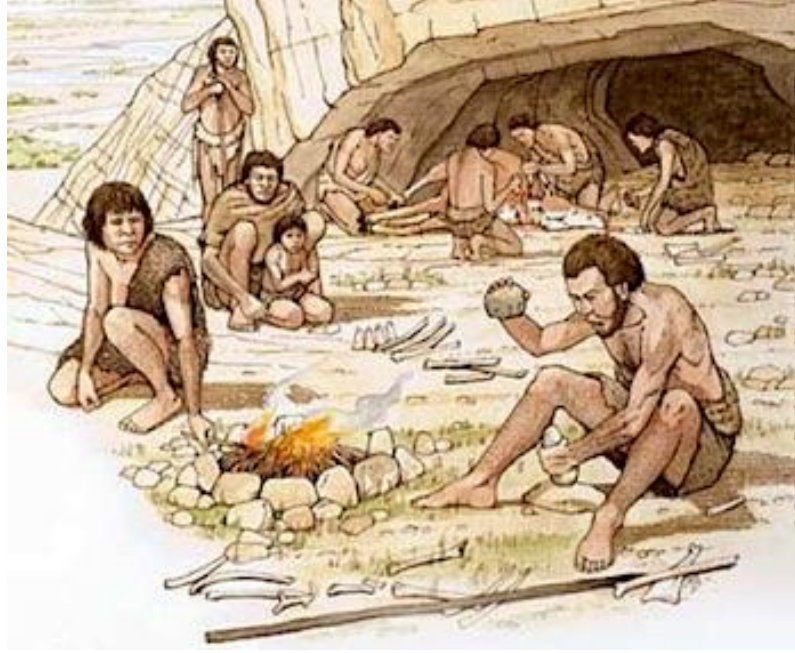
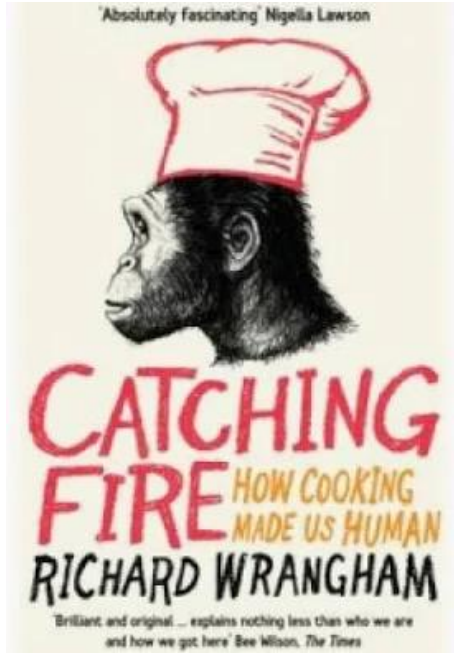


separa l'uomo dal mondo animale



Claude Lévi-Strauss
(1908 – 2009)

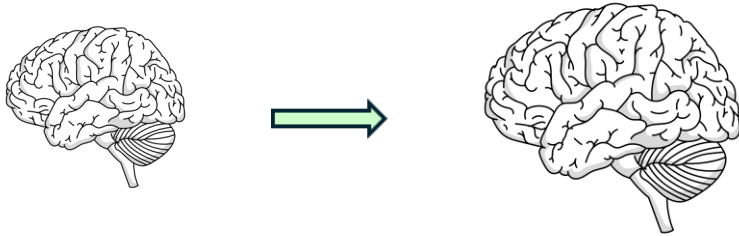
Cucinare rende umani



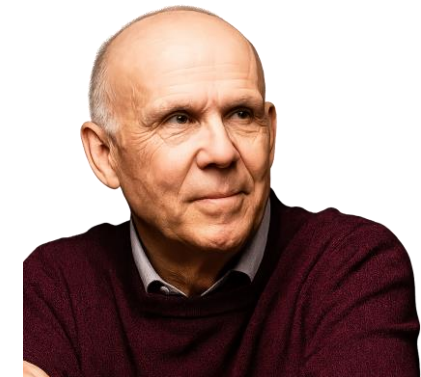
Friedrich Engels
(1820-1895)



Cottura separa l'uomo dall'animale



Cucinare → tappa decisiva dell'umanizzazione



Richard Wrangham (1948)

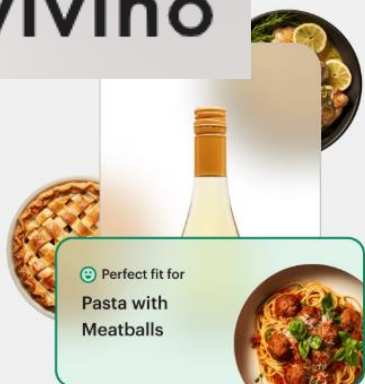
Gastronomia Computazionale

Studio computazionale della gastronomia



Applicazioni

VIVINO



Find perfect food pairings

Pair like a sommelier with recommendations for any wine you scan or search.

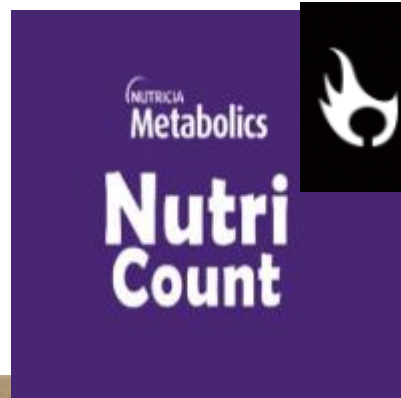


Plant Jammer

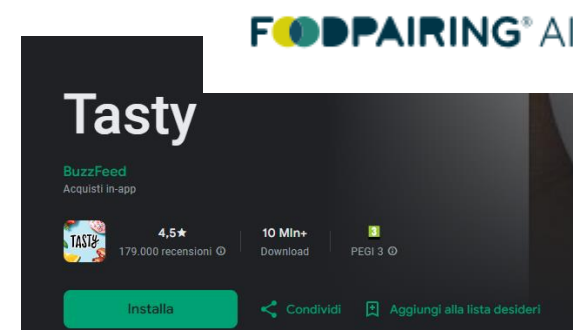
dishgen

Cognitive Cooking with Chef Watson

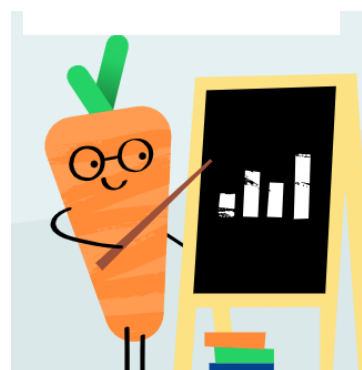
Recipes for Innovation from IBM & the Institute of Culinary Education



Perception Defined



Yuka



Wine AI

by COMMED I A

SaaS

Get it now



cookpadblog

Introducing

Cookpad's

AI Recipe Assistant

Food Pairing

‘Ci sono principi guida nella scelta degli ingredienti nelle ricette?’

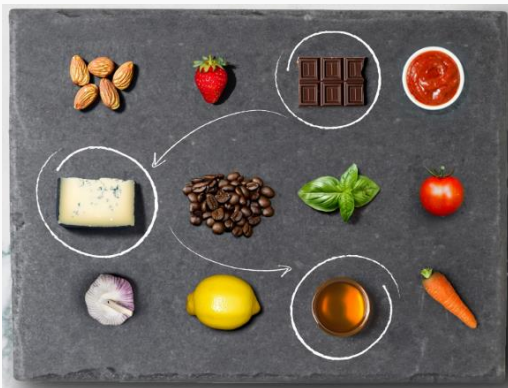


The Fat Duck
heston blumenthal

Heston Blumenthal (2008) propone una ipotesi: *‘ingredienti di sapore simile tendono ad essere associati nelle ricette’*.

FOODPAIRING® AI

<https://www.foodpairing.com/>



Basi scientifiche



Food Pairing

Flavor network and the principles of food pairing (2011)

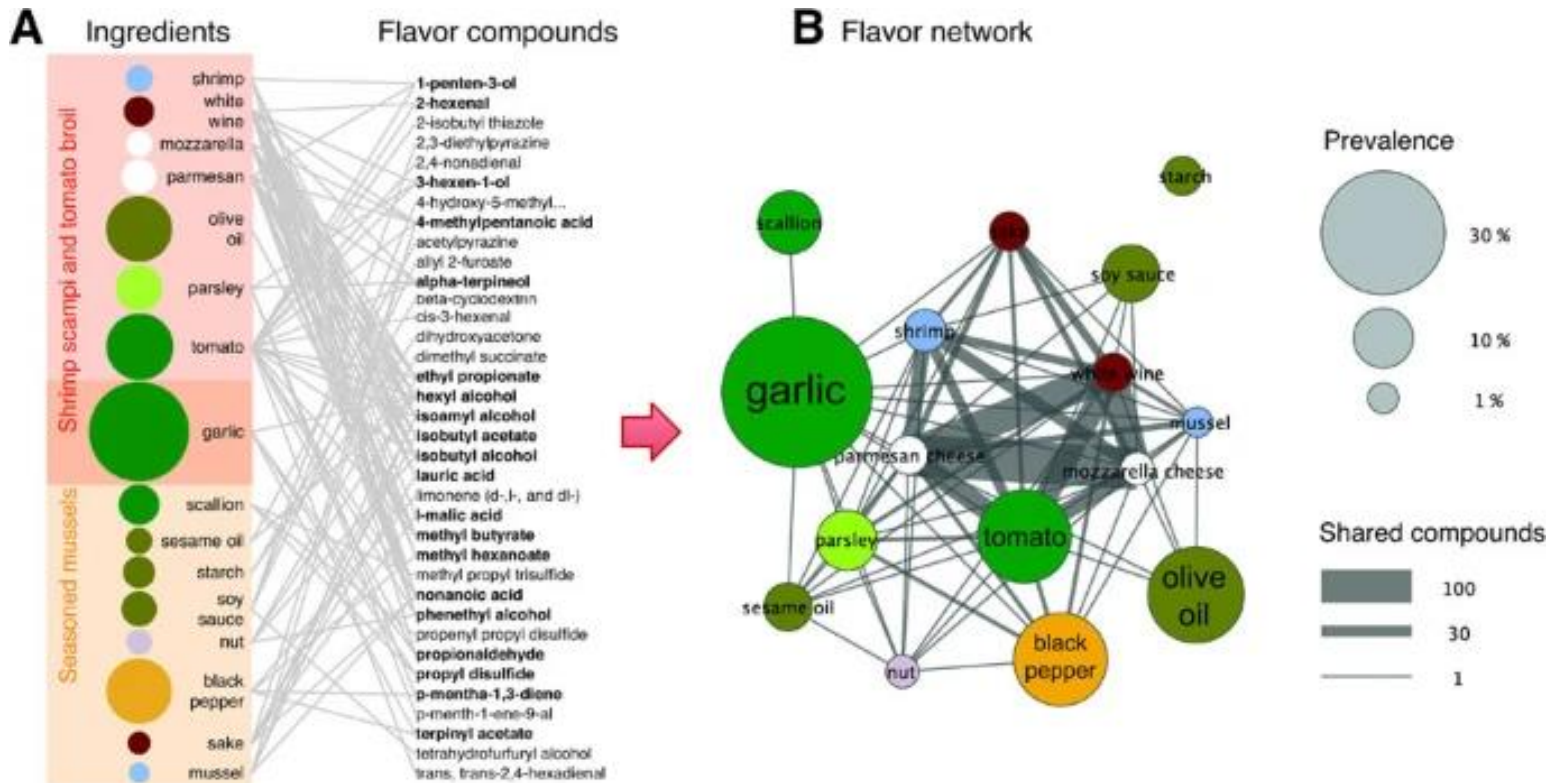
Yong-Yeol Ahn^{1,2,3*}, Sebastian E. Ahnert^{1,4*}, James P. Bagrow^{1,2} & Albert-László Barabási^{1,2}

Database Online

epicurious

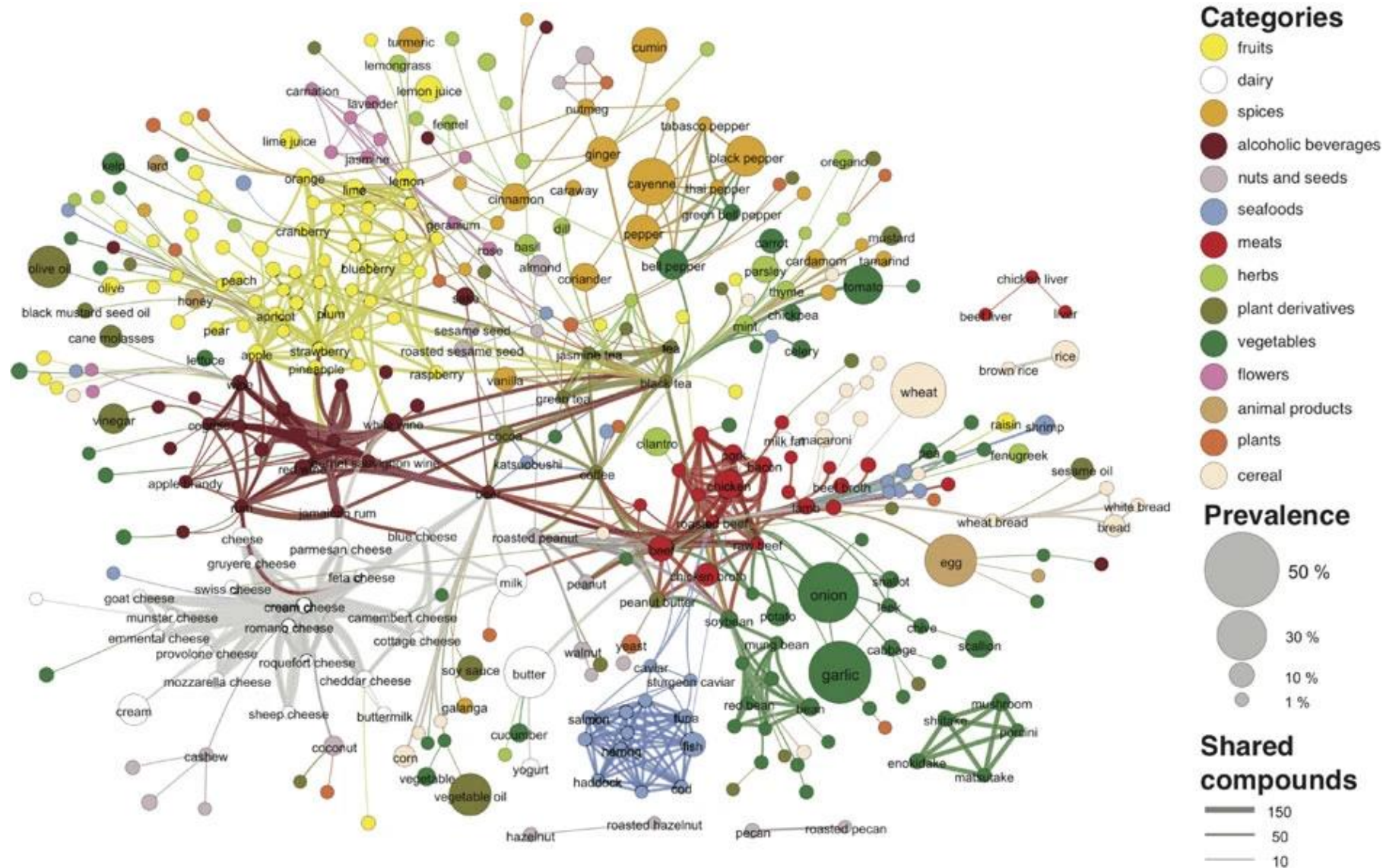
allrecipes

매일매일맛있게
Menupan



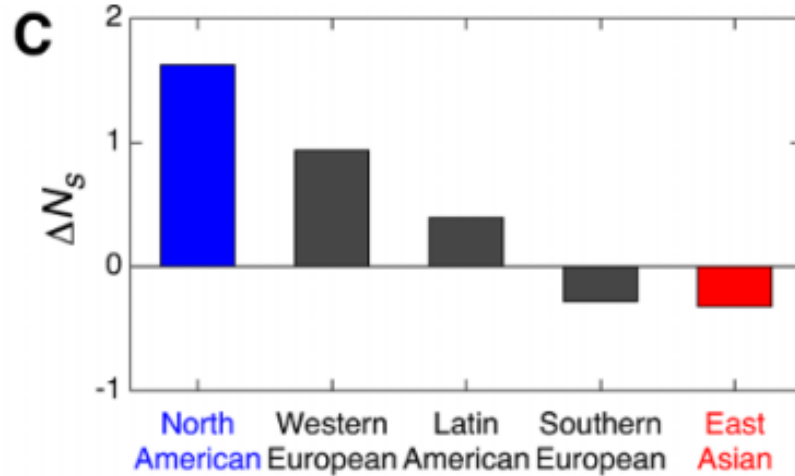
56,498 Recipes

Rete Aromatica



Food Pairing

Ingredienti di sapore simile tendono ad essere associati nelle ricette?



Modello nullo

$$\Delta N_s = N_s^{real} - N_s^{rand}$$

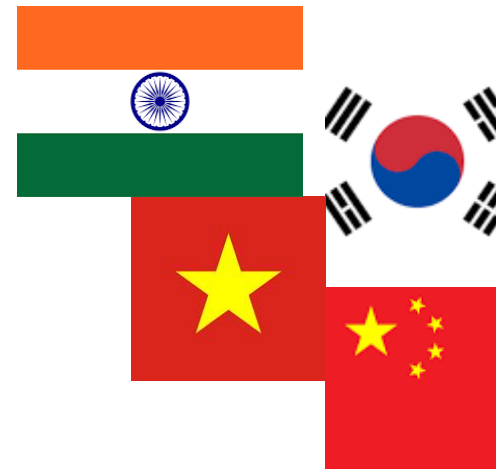
N_s^{real} : Composti condivisi ricette reali

N_s^{rand} : Composti condivisi ricette casuali



Food pairing

accoppia ingredienti
dal sapore simile

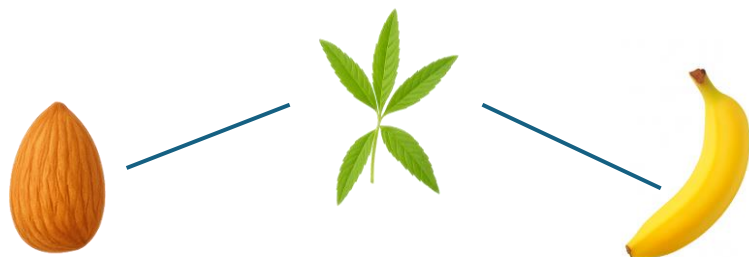


Food pairing

accoppia ingredienti
dal sapore diverso

Food Bridging

Ingredienti ponte

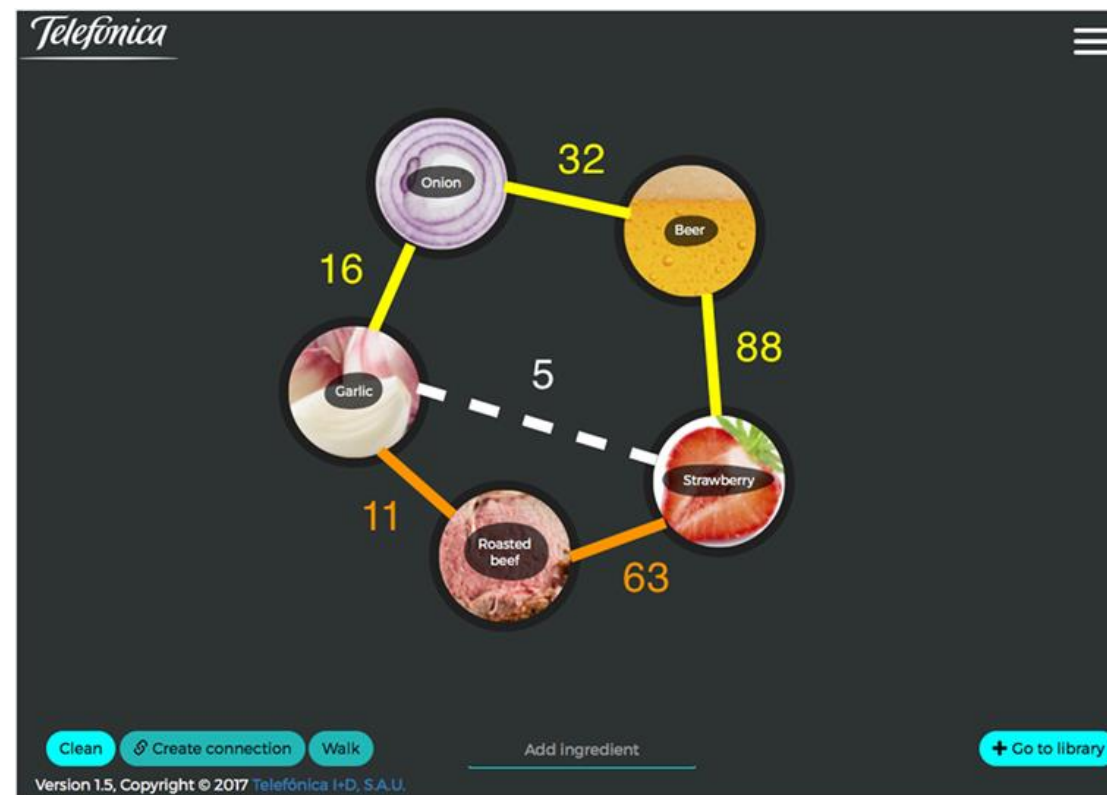


Verbena $\Rightarrow$ ingrediente ponte

Ingrediente armonizzante

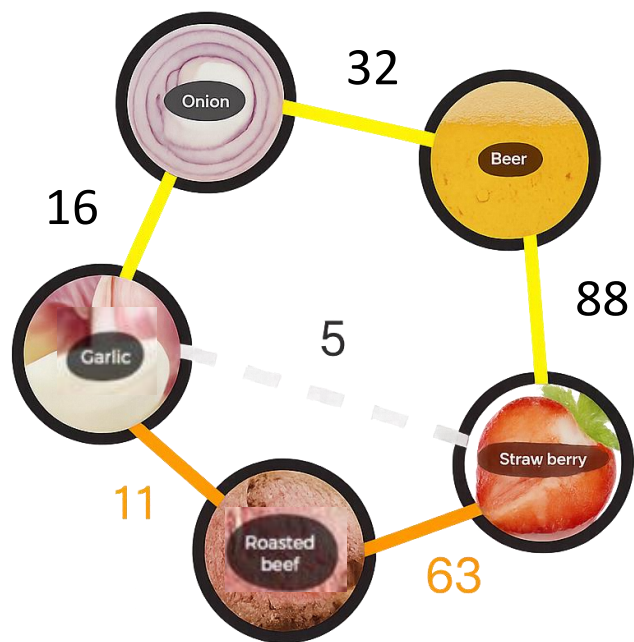
Food-Bridging: A New Network Construction to Unveil the Principles of Cooking (2013)

Tiago Simas^{1,2*}, Michal Ficek¹, Albert Diaz-Guilera^{3,4}, Pere Obrador¹ and Pablo R. Rodriguez¹



Food Bridging

metrica $\longleftrightarrow$ semi-metrica



Distanza



inverso della similarità

$d_{i,j}$ distanza diretta $\longrightarrow d_{i,j} = 1/5$

$d_{i,j}^c$ lunghezza del percorso più corto

$$d_{i,j}^c = \frac{1}{88} + \frac{1}{32} + \frac{1}{16} = 0.11$$

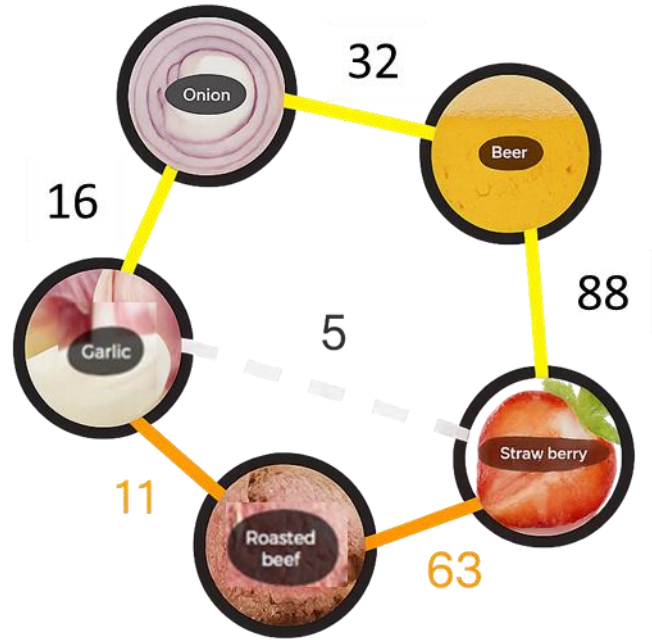
$$s_{i,j} = \frac{d_{i,j}}{d_{i,j}^c} \quad \frac{\text{Distanza diretta}}{\text{Distanza}}$$

$$s_{i,j} > 1$$

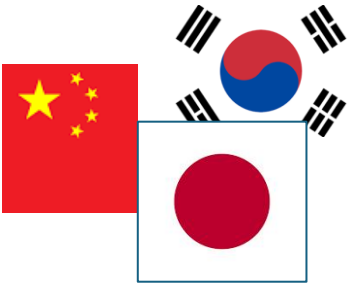


connessione semi-metrica $\rightarrow$ ingredienti ponte

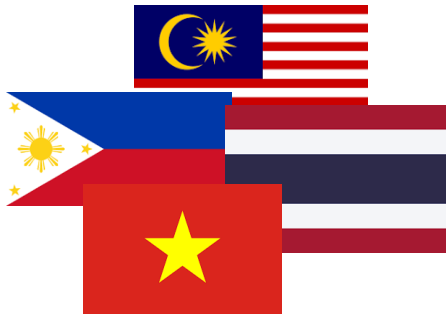
Food Bridging



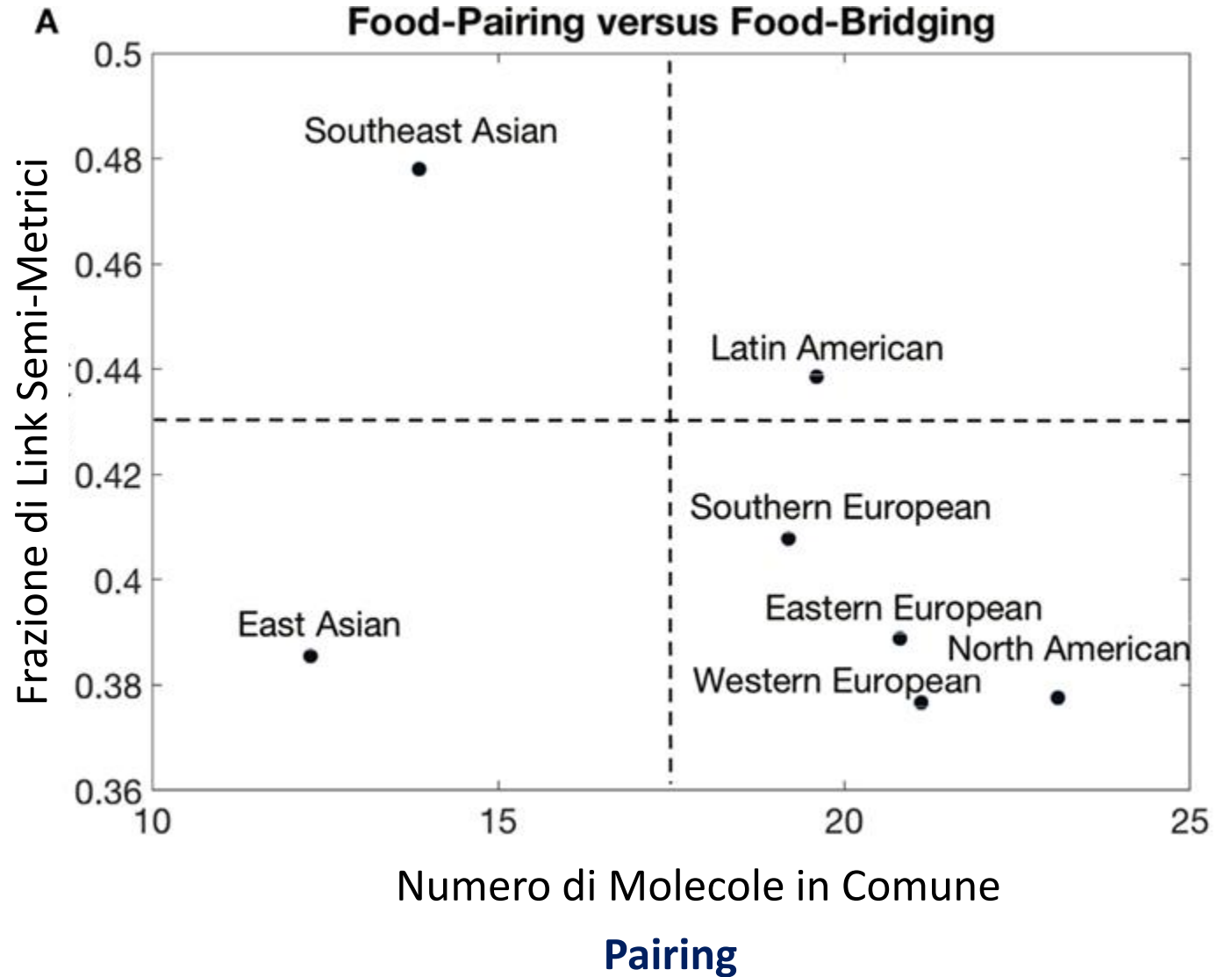
East



South East



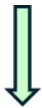
Bridging



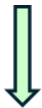
Molecole Aromatiche



Molecole aromatiche dominanti



100 volte **OAV (Odor Activity Value)**



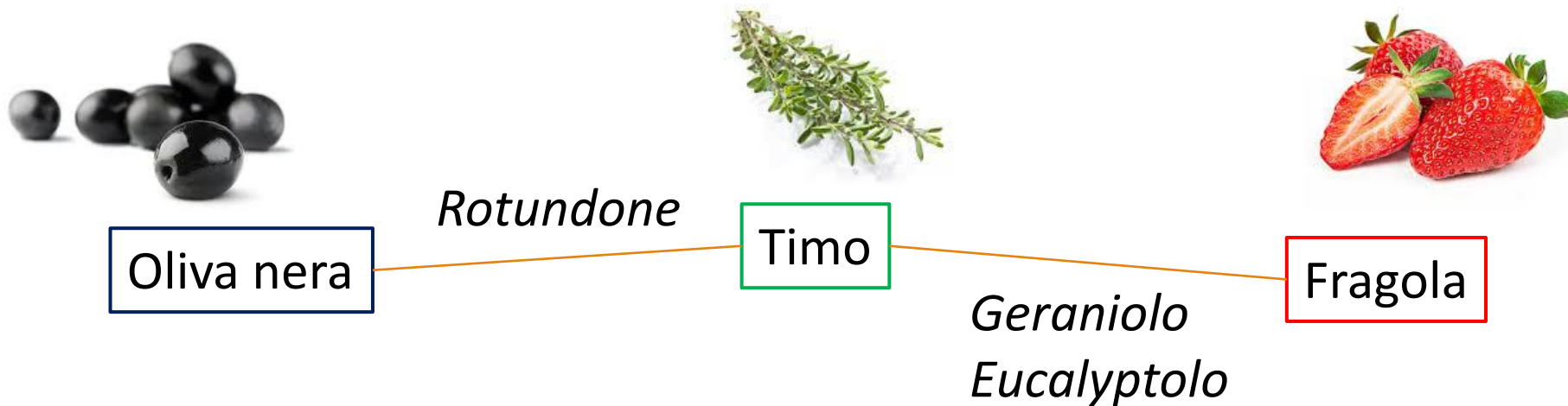
100 X soglia di percezione

Ingredienti	Molecole
balsamic vinegar	Pentanone
balsamic vinegar	Pentenal
balsamic vinegar	Propanoate d'héthyle
balsamic vinegar	Sotolon
balsamic vinegar	Syringol
balsamic vinegar	Vanilline
balsamic vinegar	5-methyl-furfural
banana	Acétate d'isoamyle
banana	Esters fruités
banana	Hexanoate d'éthyle/Isohexanoate d'e
banana	Méthanoate de Butyle
banana	Methanoate de Linalyle
banana	Octanoate d'éthyle
banana	3-Hexanol
banana	Acétate d'éthyle
banana	Butanoate d'éthyle

Network degli Aromi

Nodi  Ingredienti

Links  Presenza di molecole comuni



Molecole in Comune

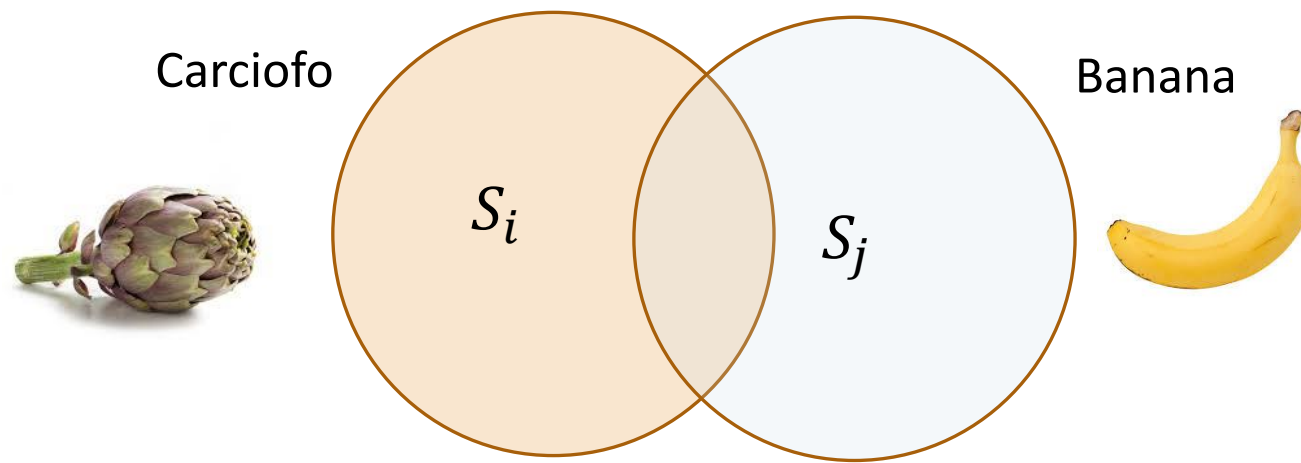
Peso dei legami

$$w_{ij} = \frac{1}{2} \left(\frac{N_{\cap}}{N_i} + \frac{N_{\cap}}{N_j} \right)$$

N_i numero molecole in S_i

N_j numero molecole in S_j

$N_{\cap}$ numero di molecole in intersezione, i.e. $|S_i \cap S_j|$

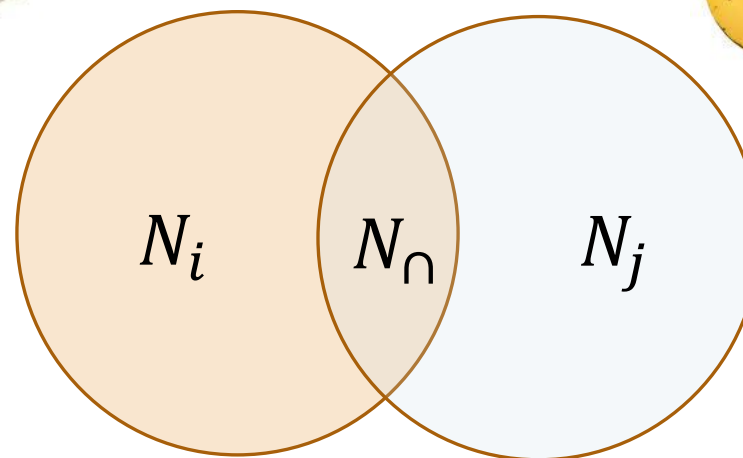


Molecole in Comune

Peso dei links

$$w_{ij} = \frac{1}{2} \left(\frac{N_{\cap}}{N_i} + \frac{N_{\cap}}{N_j} \right)$$

$$w_{ij} \in [0,1]$$



Alta intersezione



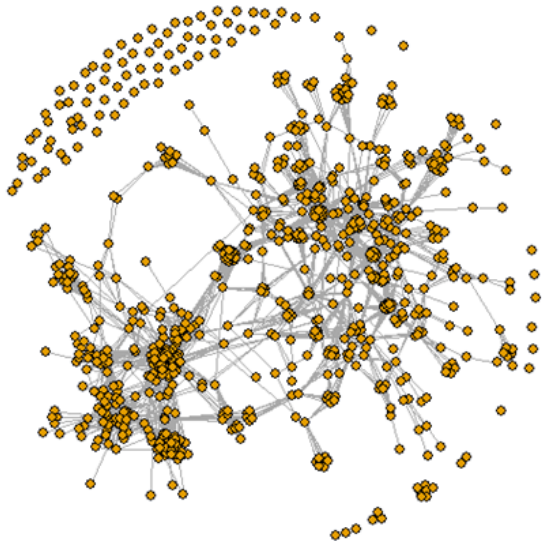
Alta similarità



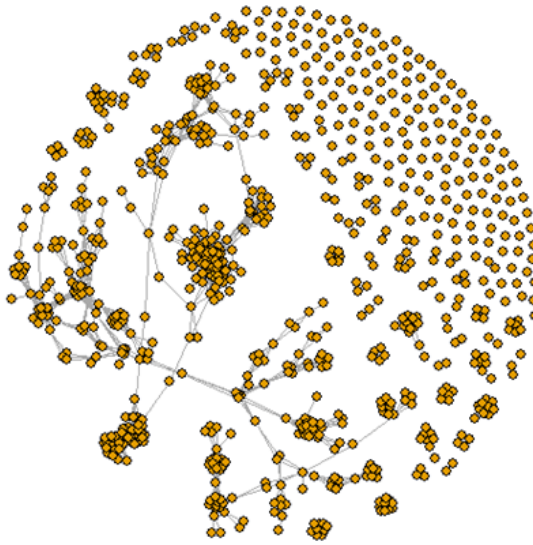
Alto peso dei links

Network di Crescente Similarità

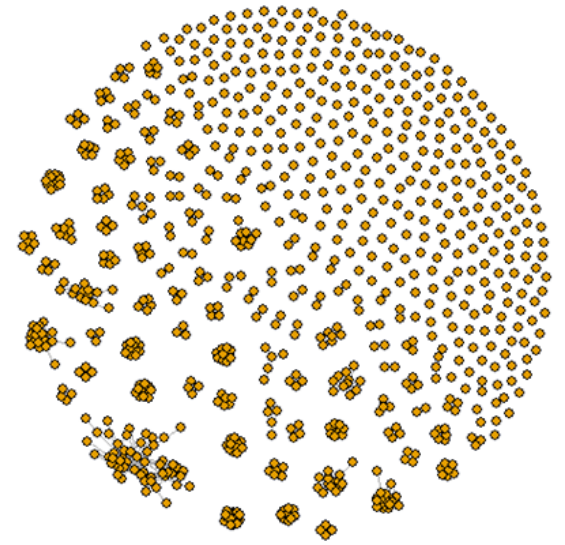
$th=0.6$



$th=0.7$

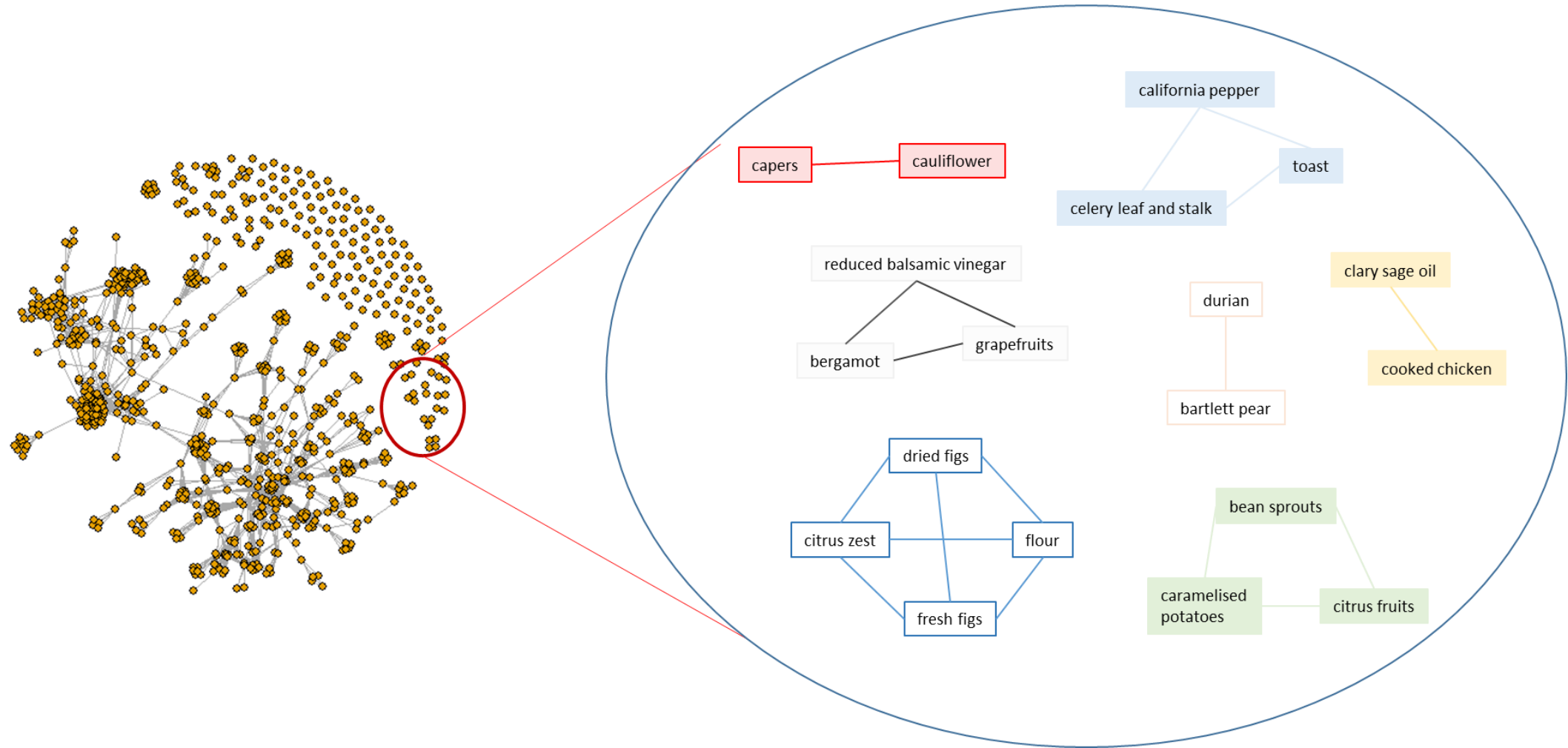


$th=0.8$



Ingredienti connessi sono più simili

Comunità di Ingredienti



Ingredienti Ponte

Algoritmo

Scopre ingredienti capaci di connettere ingredienti non connessi



Pesco

Ponte



Salmone

Ingredienti Ponte

Algoritmo

Scopre ingredienti capaci di connettere ingredienti non connessi



Network delle Ricette

Nodi



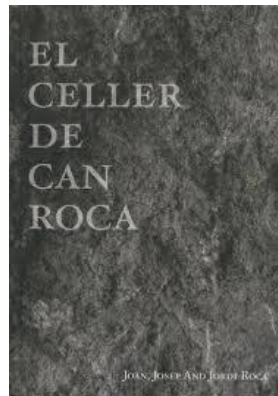
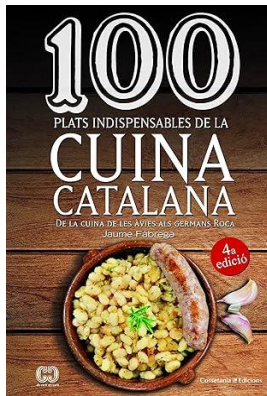
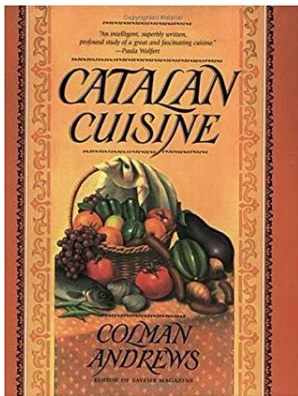
Ricette

Links



Ingredienti in Comune

Ricettari



scientific reports (2025)

The recipe similarity network: a new algorithm to extract relevant information from cookbooks

Michele Bellingeri^{1,2}, Axel Bidon-Chanal Badia³, Marta Vila Rigat⁴, Roberto Alfieri^{1,2}, Massimiliano Turchetto^{1,2} & Davide Cassi^{1,2}



UNIVERSITAT DE
BARCELONA

Ingredienti in Comune

??

=

Struttura Comune



Ontologia dei Dati

B	C	D	E	F
TEXT COMPLET	Recipe ID	Base	Lema Base	Paràfrasis Base
1 ànec mut	61	ànec	ànec	ànec
1 bitxo	72	bitxo	bitxo	bitxo
1 bitxo	47	bitxo	bitxo	bitxo
1 bola de sagí	17	sagí	sagí	sagí
1 bola de sagí enfarinada	14	sagí	sagí	sagí
1 botifarra crua	58	botifarra	botifarra	botifarra
1 botifarra crua per persona	23	botifarra	botifarra	botifarra
1 branqueta de farigola	98	farigola	farigola	farigola
1 branqueta de julivert	28	julivert	julivert	julivert
1 cabdell de col	65	col	col	col
1 cabeça d'alls escalivada	71	alls	all	all
1 cabeça d'alls	4	alls	all	all
1 cabeça d'alls	16	alls	all	all
1 cabeça d'alls	24	alls	all	all
1 cabeça d'alls	33	alls	all	all
1 cabeça d'alls	55	alls	all	all
1 cabeça d'alls	57	alls	all	all
1 cabeça d'alls	63	alls	all	all
1 cabeça d'alls	66	alls	all	all
1 cabeça d'alls	68	alls	all	all
1 cabeça d'alls	94	alls	all	all
1 cabra d'uns 2 kg	29	cabra	cabra	cabra
1 cafè acabat de fer	89	cafè	cafè	cafè
1 canó de canyella	76	canyella	canyella	canyella
1 canó de canyella	85	canyella	canyella	canyella
1 carbassó petit	97	carbassó	carbassó	carbassó
1 ceba	9	ceba	ceba	ceba

{cipolla, cipolla bianca, cipolla rossa, cipolla rossa a pezzi}

Cipolla



Insalata di pollo con maionese

{Pollo, **Maionese**, Sedano, Sale, Pepe, Limone, Lattuga}



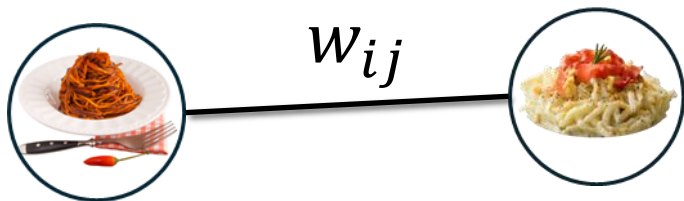
{Uova, Olio, Limone, Sale}

Sostituzione

Insalata di pollo con maionese

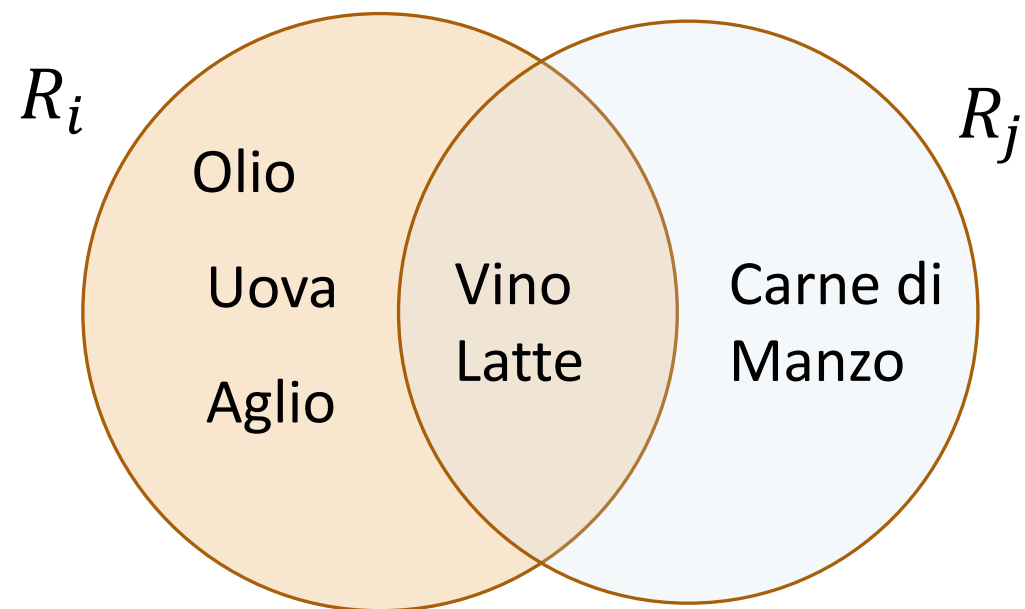
{Pollo, Uova, Olio, Limone, Sale, Sedano, Pepe, Lattuga}

Similarità tra Ricette



Peso dei Links:

$$w_{ij} = \frac{1}{2} \left(\frac{N_{\cap}}{N_i} + \frac{N_{\cap}}{N_j} \right)$$



Alta sovrapposizione $\Rightarrow$ Elevata Similarità $\Rightarrow$ Elevato Peso Links

Costruzione del Network

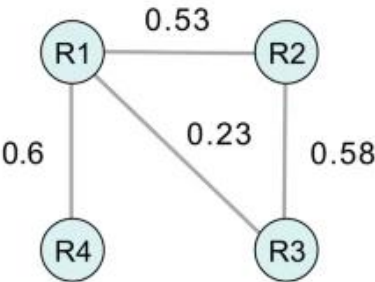


Nodi $\Rightarrow$ Ricette

Links $\Rightarrow$ Ingredienti Comuni

A	Recipes	B	Recipes Intersection	C	Intersection Graph	D	Trivial Weighted Network
	R1: {Oil, Wine, Pepper, Egg, Garlic}		R1 $\cap$ R2: {Wine, Pepper}				
	R2: {Wine, Pepper, Meat}		R1 $\cap$ R3: {Wine}				
	R3: {Wine, Meat, Onion, Celery}		R1 $\cap$ R4: {Oil, Egg, Garlic}				
	R4: {Oil, Wine, Fish, Egg, Garlic}		R2 $\cap$ R3: {Wine, Meat}				
			R2 $\cap$ R4: $\{\emptyset\}$				
			R3 $\cap$ R4: $\{\emptyset\}$				

E Recipe Similarity Network

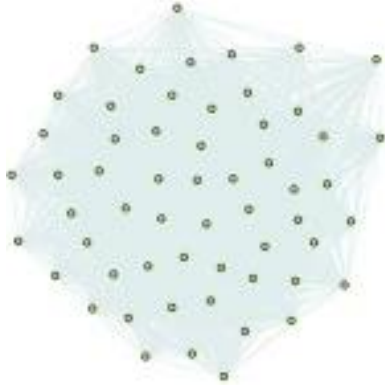


Rimozione di links

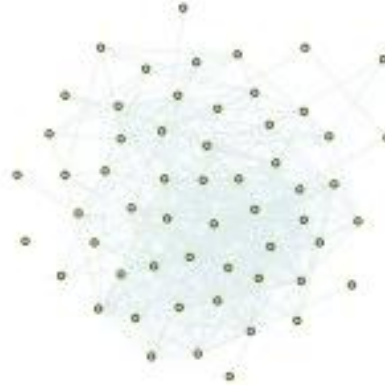
EL
CELLER
DE
CAN
ROCA

JOSÉ JUAN AND JUAN ROCA

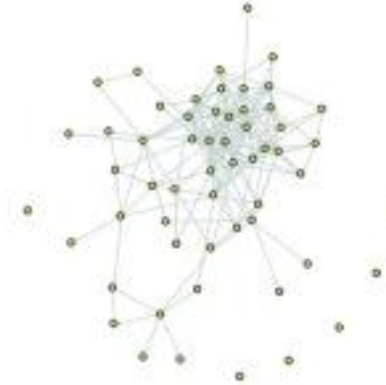
$w > 0$



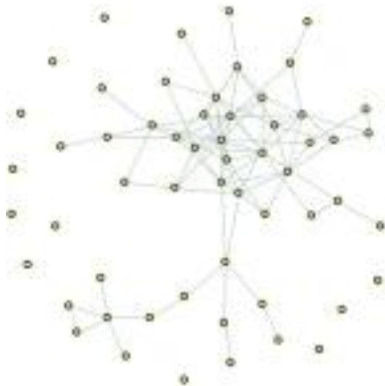
$w > 0.40$



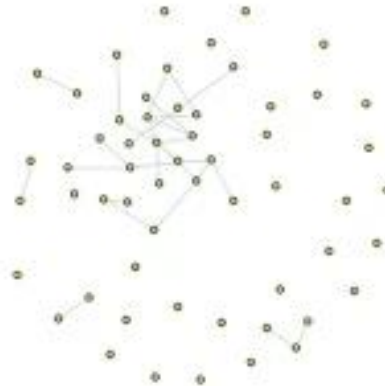
$w > 0.45$



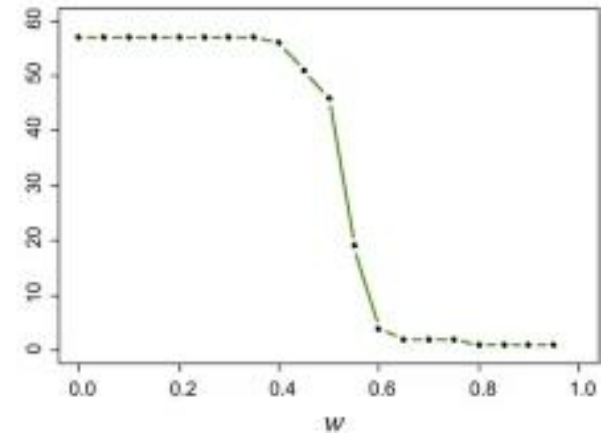
$w > 0.50$



$w > 0.55$

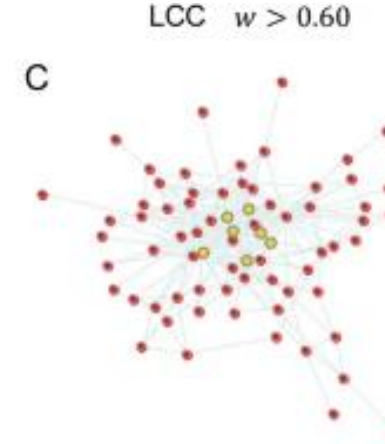
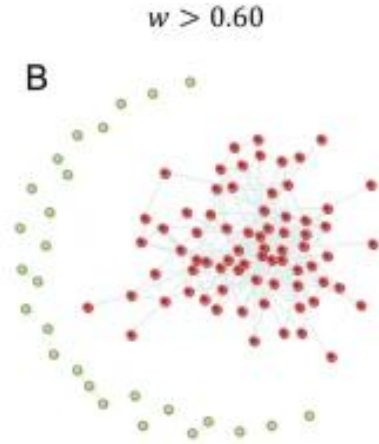
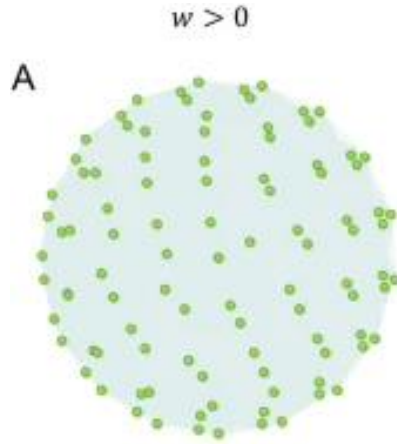


LCC

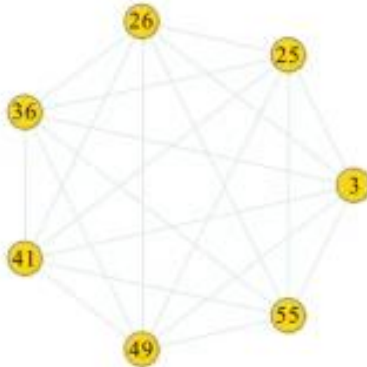


Clique detection

max_cliques() R igraph, Bron–Kerbosch con pivoting
(Tomita et al., 2006)



D 7-clique $w > 0.60$



E

Node 3 : *Allioli Sauce*

Node 25 : *Bean and Potato Stew*

Node 26 : *Mixed meat paella*

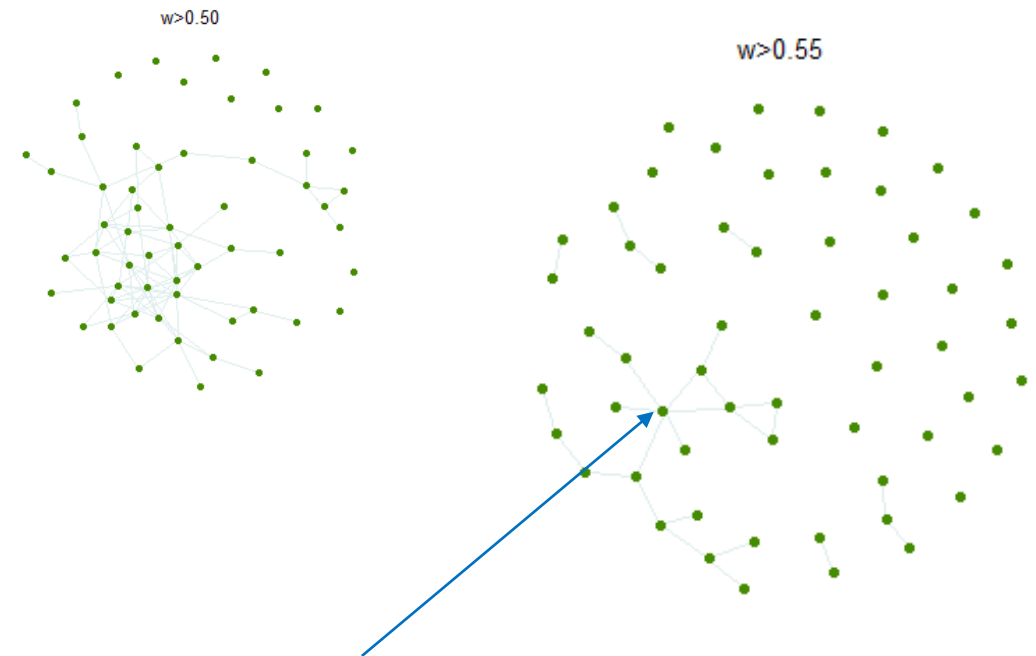
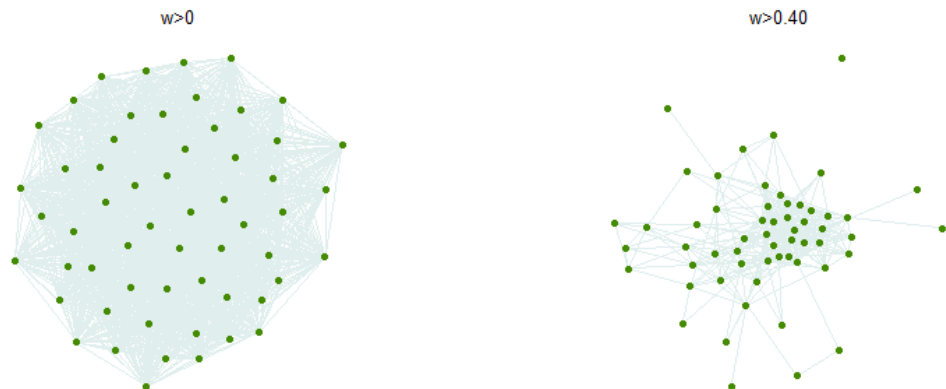
Node 36 : *Fairy rings mushrooms omelette*

Node 41 : *Monkfish Stew*

Node 49 : *Grilled Langoustines*

Node 55 : *Dry Salted Codfish Soup*

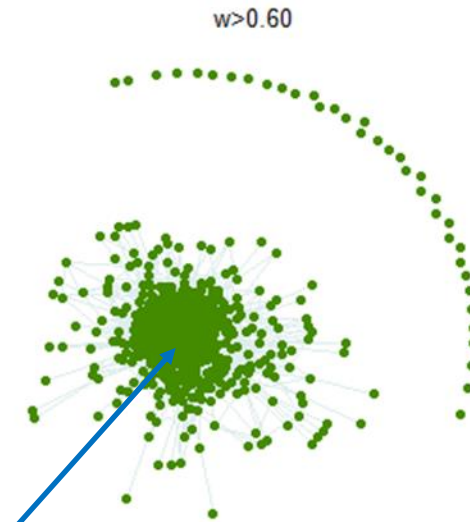
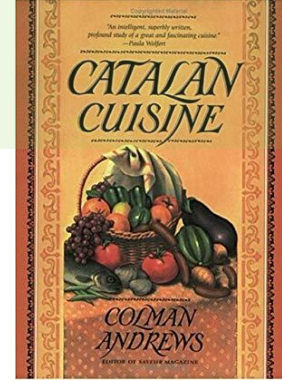
A wooden sign with a stylized 'R' logo above it, reading "EL CELLER DE CAN ROCA". The sign is mounted on a wooden wall.



Saigua, becada, bolet, ceba, farina, flor de romaní, kuzu,
llet de vaca, llevat, mantega, oli, ou de gallina, pastanaga,
porro, sal, sucre, tòfona

[illegible]

Ricette Caratterizzanti



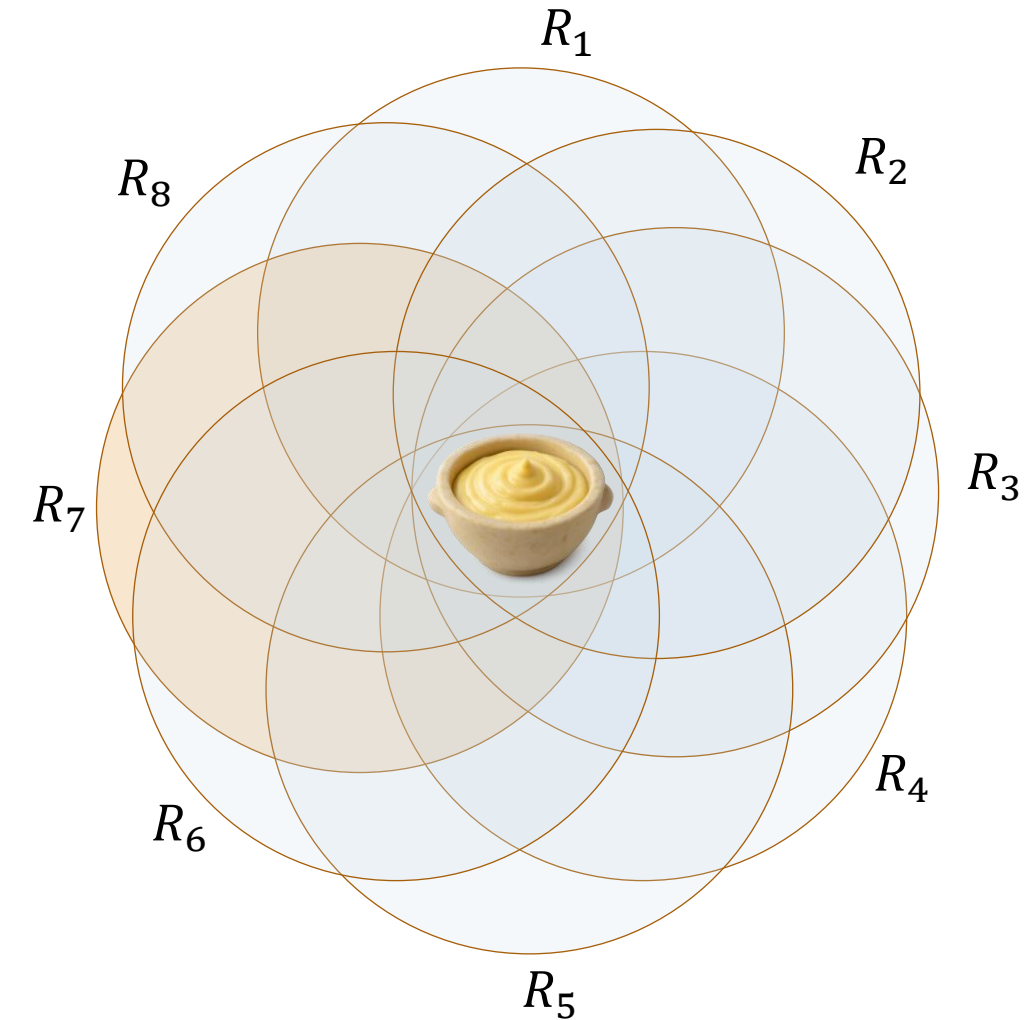
Allioli

Salt, Garlic, Olive oil



Ricette caratterizzanti

Tradizionale $\Rightarrow$ *'Ricetta Fondante'*



Haute $\Rightarrow$ *'Ricetta Manifesto'*

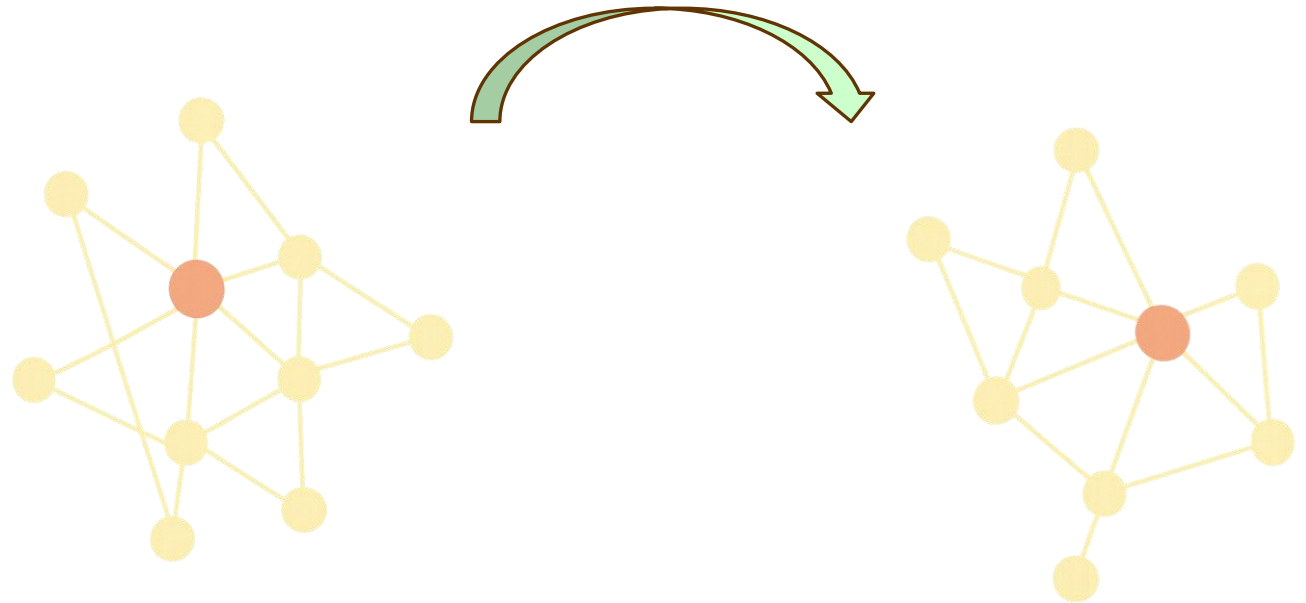


Ricette caratterizzanti

10 ricette caratterizzanti per il Ctrad

Rank	No Substitution		Substitution	
	Recipe ID	s	Recipe ID	s
1	144	311.62	55	327.30
2	962	311.62	144	327.30
3	124	304.9	133	311.67
4	1063	301.3	1063	311.54
5	1049	300.81	1122	305.60
6	1066	300.61	542	303.77
7	859	300.55	949	301.89
8	803	297.94	20	300.51
9	564	297	984	300.05
10	133	296.29	124	299.05

Sostituzione



Ontologia dei Dati ➡ Struttura della rete

Gastronomia computazionale

Venti ingredienti più frequenti

Ctrad		101		Roca	
Ingredients	Frequency	Ingredients	Frequency	Ingredients	Frequency
sal	508	sal	81	sal	55
oli	467	oli	75	aigua	53
all	320	all	46	oli	49
aigua	298	ceba	45	ceba	30
ceba	285	aigua	41	sucre	30
pebre	249	tomàquet	40	mantega	29
tomàquet	224	pebre	31	pastanaga	29
julivert	174	porc	24	goma de xantè	28
vi	130	julivert	20	porro	24
farina	128	ametlla	19	ou de gallina	23
porc	120	farina	18	nata	21
ou de gallina	109	vi	18	farina	14
patata	97	ou de gallina	16	llet de vaca	14
ametlla	96	pa	16	oli de gira-sol	14
llorer	92	sucre	12	tòfona	13
pa	87	cansalada	11	agar-agar	12
arròs	52	canyella	11	all	12
safrà	51	llorer	11	taronja	12
cansalada	49	patata	11	escalunya	11
mongetes	48	pollastre	10	fonoll	10

Data Mining

Cibo → Salute

scientific reports

OPEN

Exploring food contents in scientific literature with FoodMine (2020)

Forrest Hooton¹, Giulia Menichetti¹ & Albert-László Barabási^{1,2,3}✉



Frida fooddata.dk



United States
Department of
Agriculture

Tracciano le sostanze chimiche??



Dr. Duke's Phytochemical and Ethnobotanical Databases

Cite

Download all (5.77 MB)

Share

Embed

+ Collect

Data Mining

FoodMine  **NLP** Identifica articoli PubMed che riportano dati sulla composizione chimica degli alimenti



- ~ 70% sostanze chimiche **non presente** nelle banche dati
- potenziale rilevanza per la **salute**: metalli, tossine, pesticidi, ecc.

Banche Dati

Datasets

Search: food

All datasets X Computer Science Education Classification Computer Vision

8,100 Datasets

Food Images (Food-101)
K Scott Mader · Updated 8 years ago
Usability 8.1 · 101016 Files (other, JSON) · 6 GB · 54.4K downloads · 229 notebooks

Database Profile

FlavorDB



Wine Aroma Wheel from UC Davis



Recipe1M+: A Dataset for Learning Cross-Modal Embeddings for Cooking Recipes and Food Images



DOHMH MenuStat (Historical)

Metadata Updated: September 2, 2023

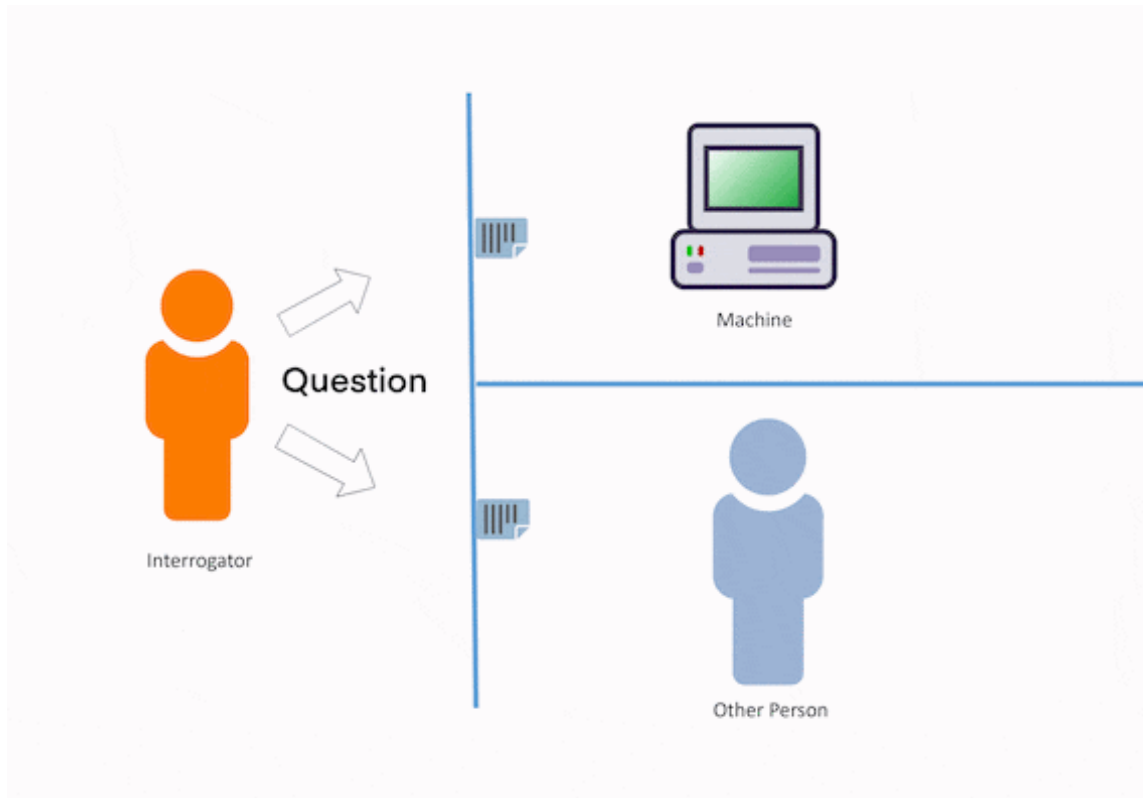


FSBI-DB

epicurious
Recipes & Menus



Can a machine think like a Chef?



Generazione Ricette Artificiali



 **Ratatouille**
Create Novel Recipes

Stuck in a dinner dilemma?
Generate novel recipes with ingredients of your choice.

- 24 chef coinvolti
- 1.472 ricette valutate

Can a machine think like a Chef?

Matrice di Confusione

	Pred. Reale	Pred. Non reale
Reale	Vero positivo	Falso negativo
Non Reale	Falso positivo	Vero negativo

$$\textit{Precisione} = \frac{n \textit{ Veri Positivi}}{n \textit{ Veri Positivi} + n \textit{ Falsi Positivi}}$$

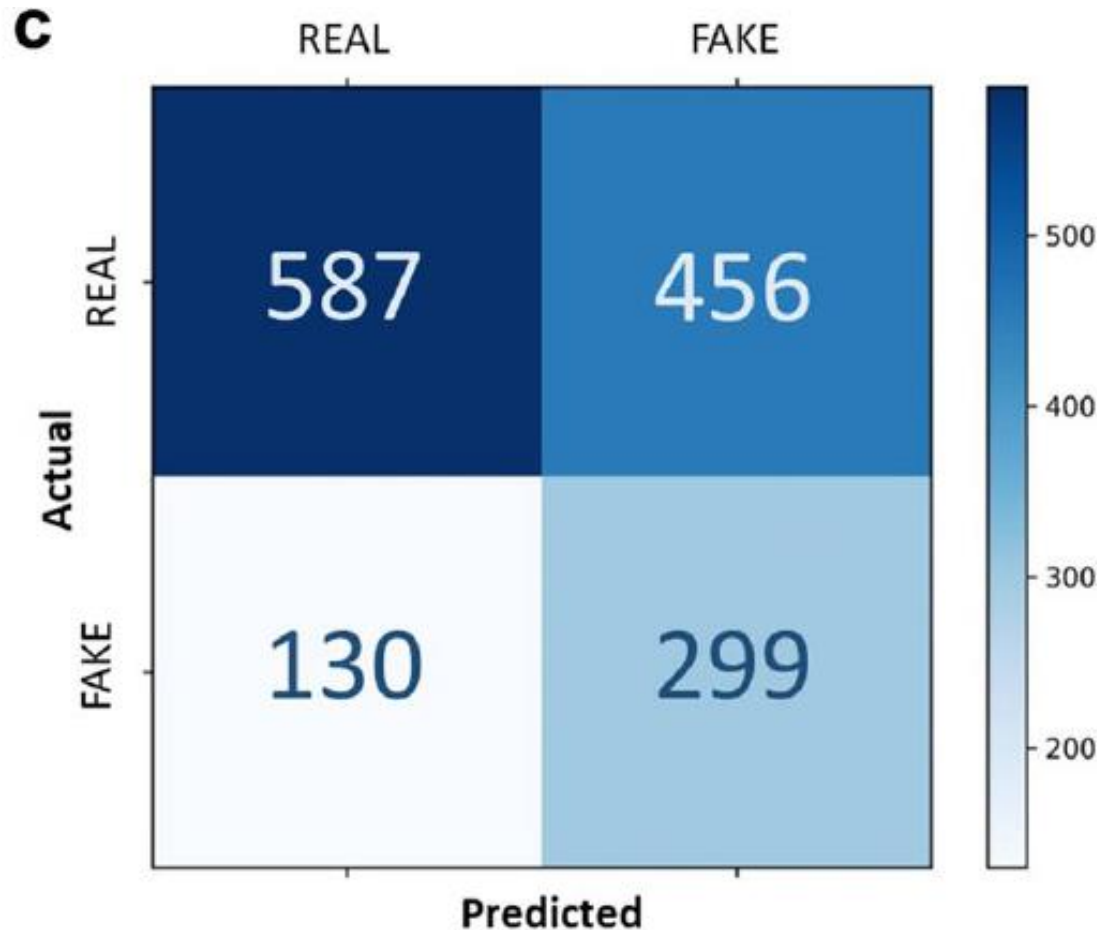
Quanto preciso nel predire ricette reali?

$$\textit{Richiamo} = \frac{n \textit{ Veri Positivi}}{n \textit{ Veri Positivi} + n \textit{ Falsi Negativi}}$$

Si lascia sfuggire ricette reali?

Can a machine think like a Chef?

Matrice di Confusione



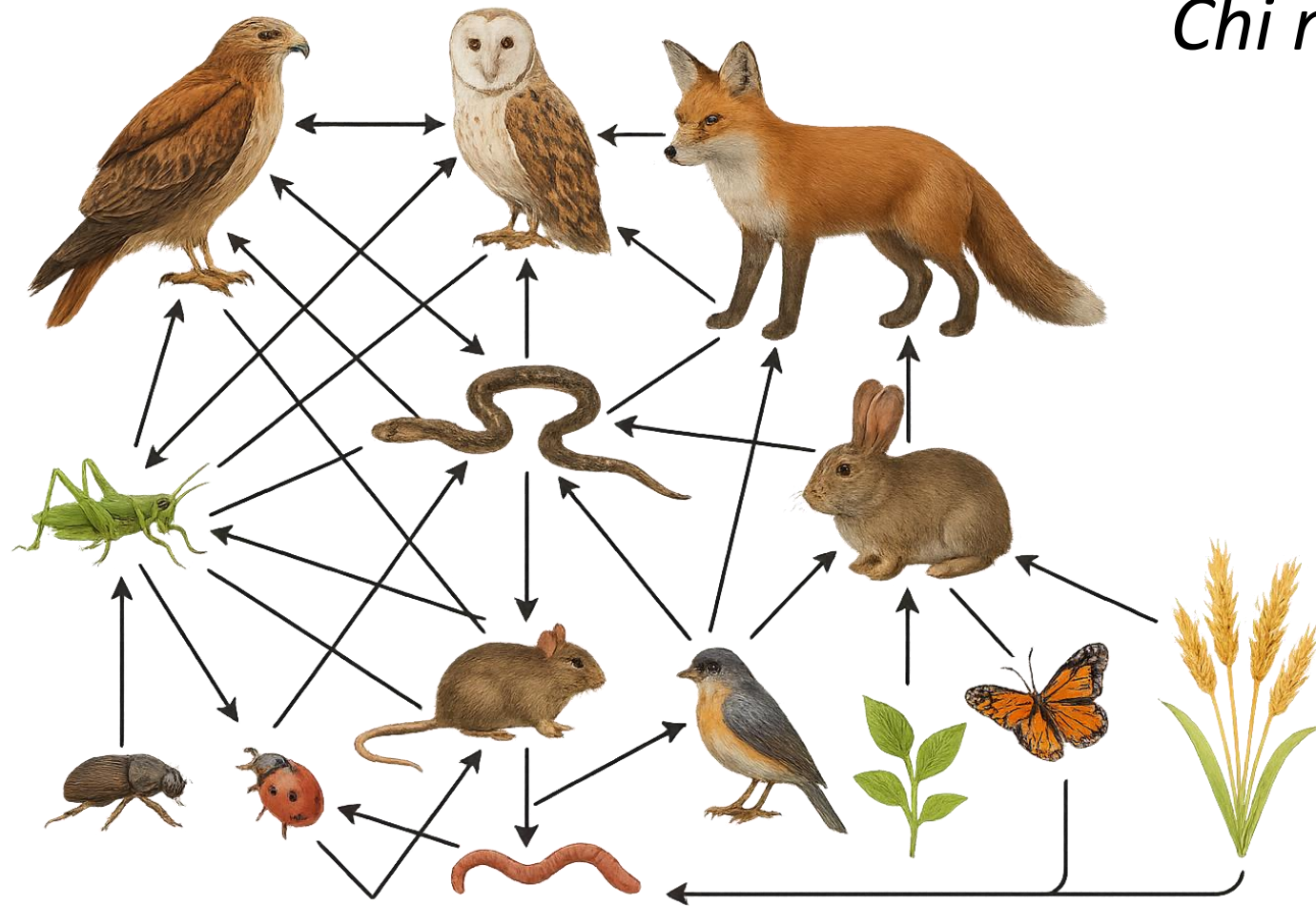
$$F1 = 2 \cdot \frac{\text{Precisione} \cdot \text{Richiamo}}{\text{Precisione} + \text{Richiamo}}$$

F1-score = 69,88%



Ricette sufficientemente credibili da
“quasi” superare il Turing Test per chef.

Food webs



Chi mangia cosa negli ecosistemi?

Complex Networks Ecologici

Specie  Nodi

Flussi Trofici  Links

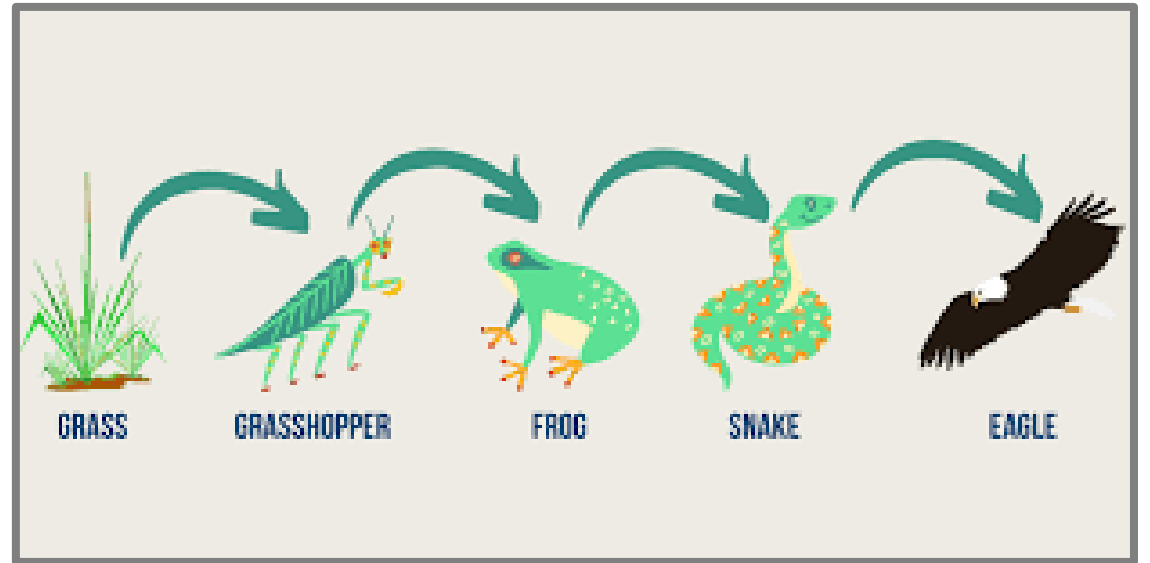
Estinzione a Domino

Singola Estinzione $\Rightarrow$ Molte Estinzioni

'Domino extinction'

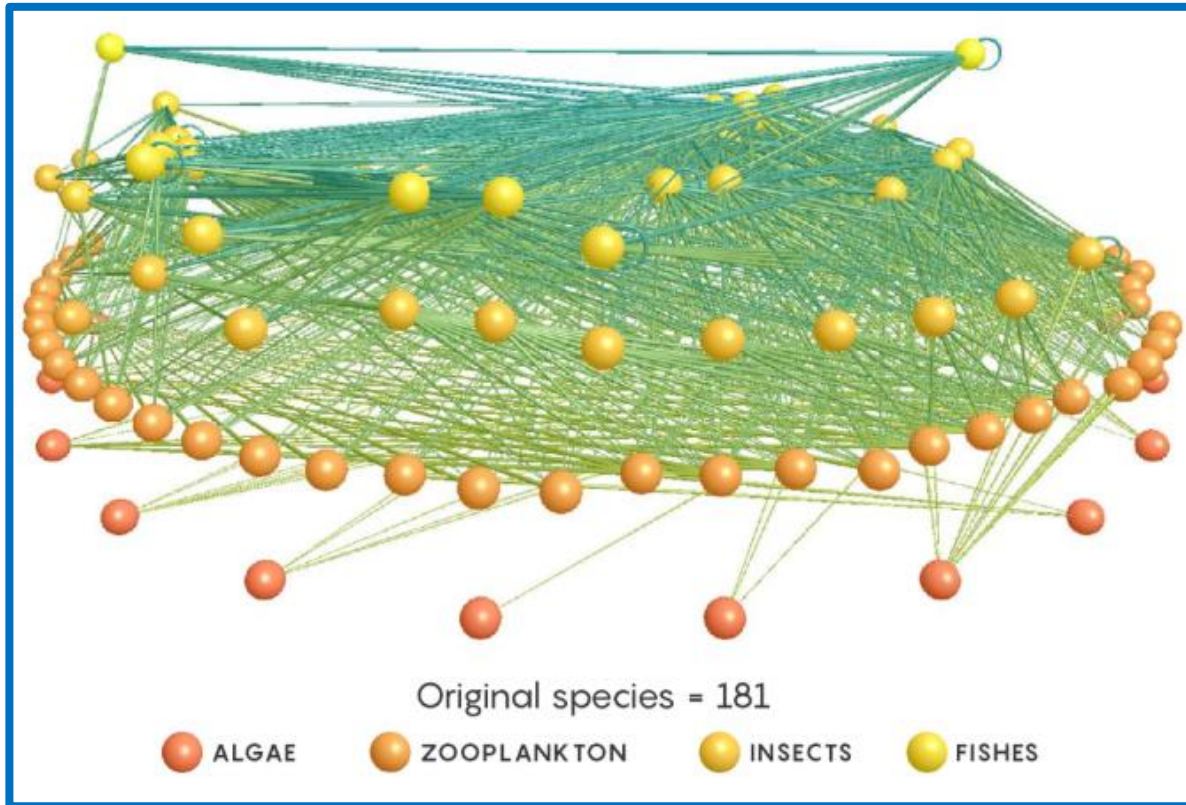


Canale Trofico



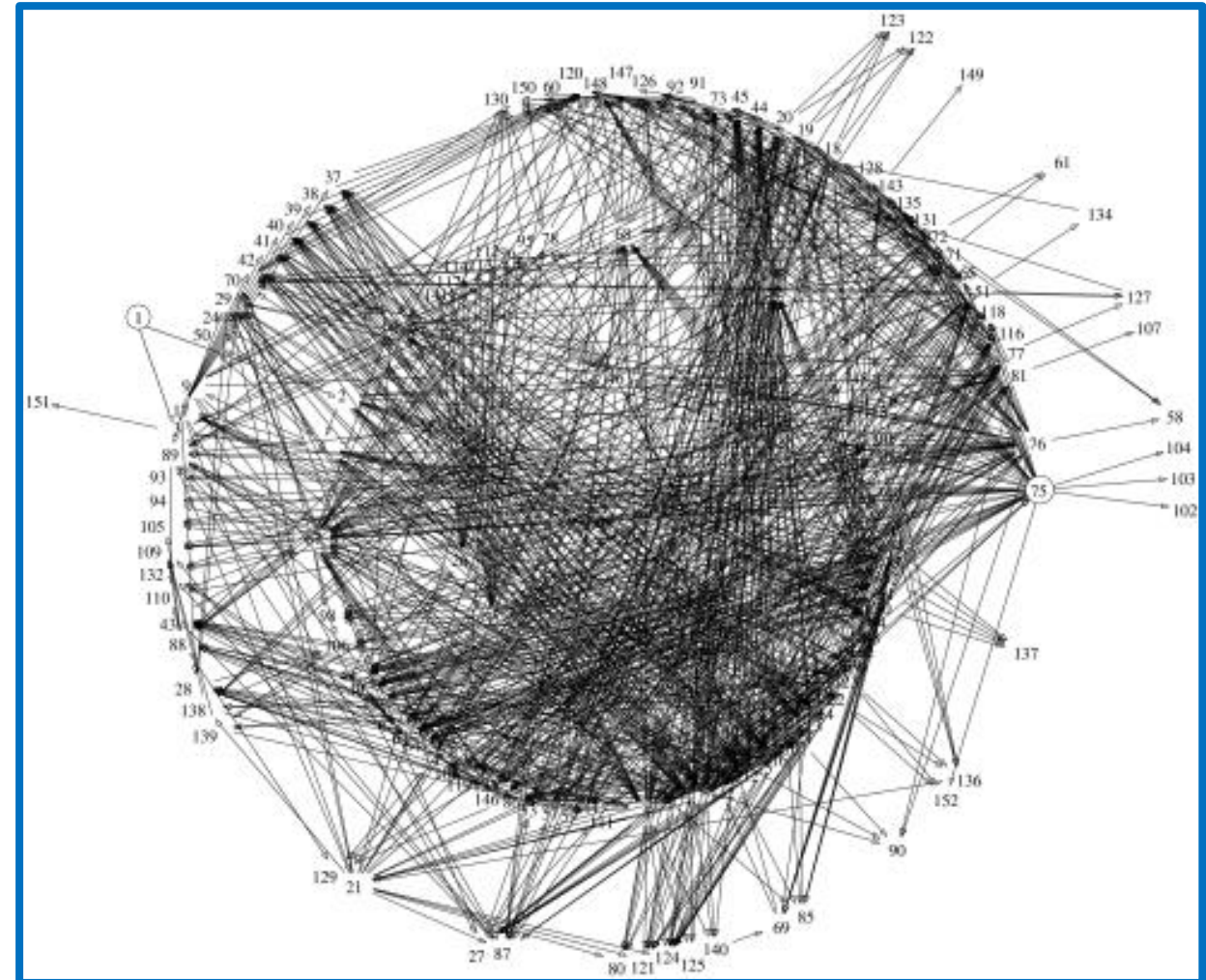
Food webs

Lago di Little Rock



[10.1145/1066677.10669](https://doi.org/10.1145/1066677.10669)

Mare di Barents

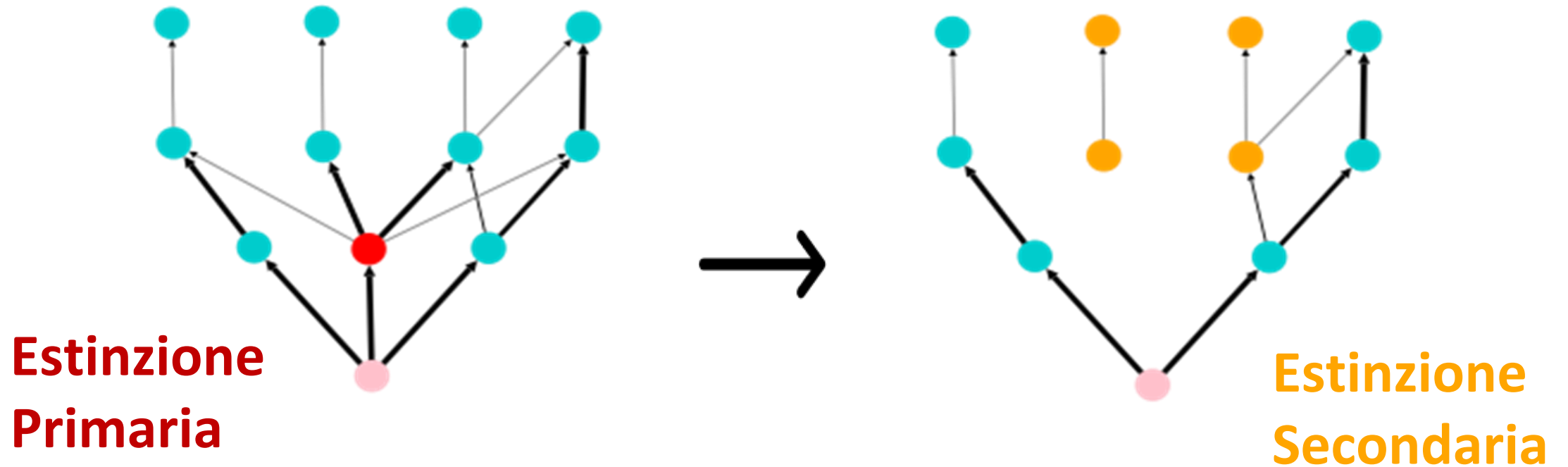


[10.1098/rstb.2008.0278](https://doi.org/10.1098/rstb.2008.0278)

Estinzione secondaria

Quante specie perdono risorsa energetica?

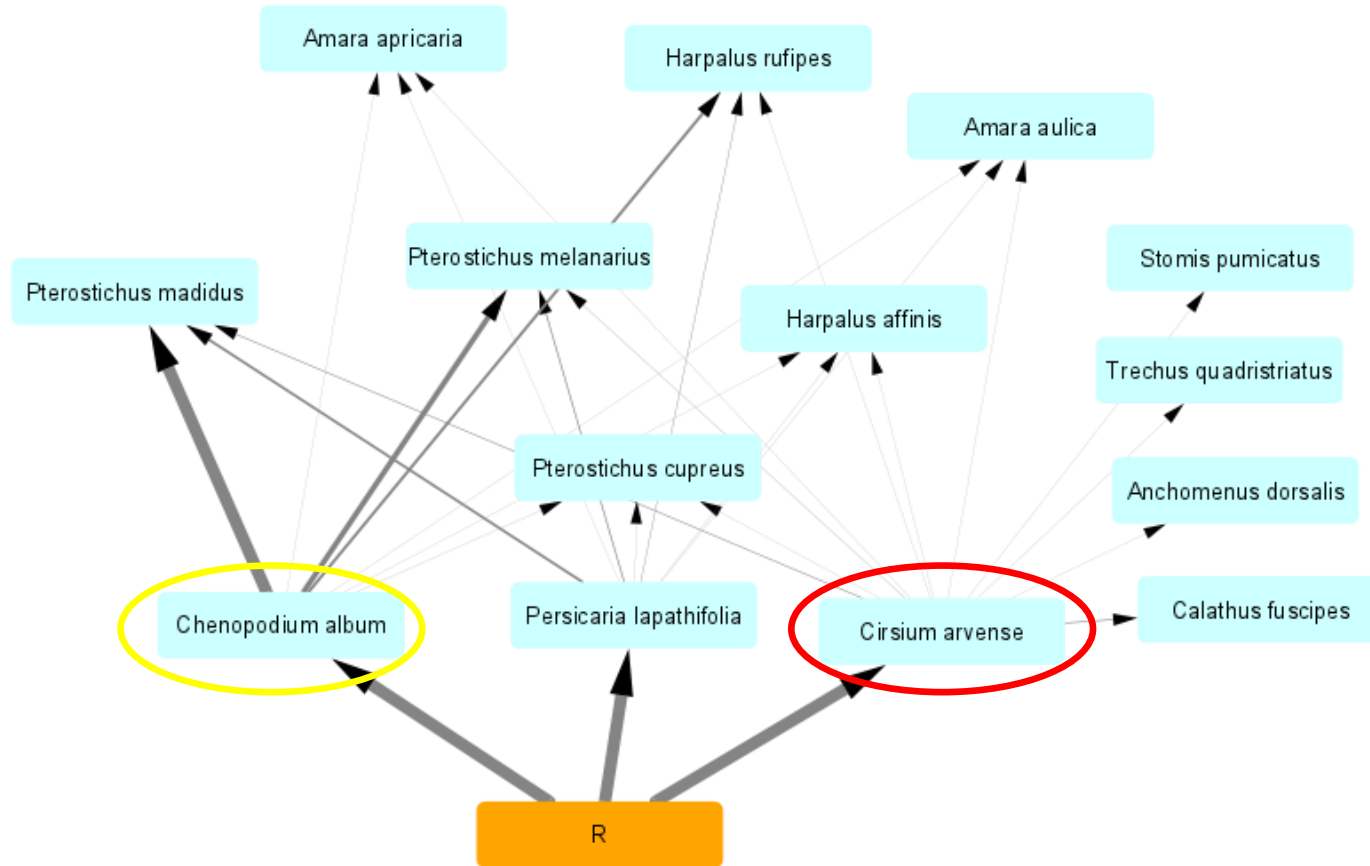
Criterio Energetico



Specie Chiave

[illegible]

Specie Chiave



Chenopodium album

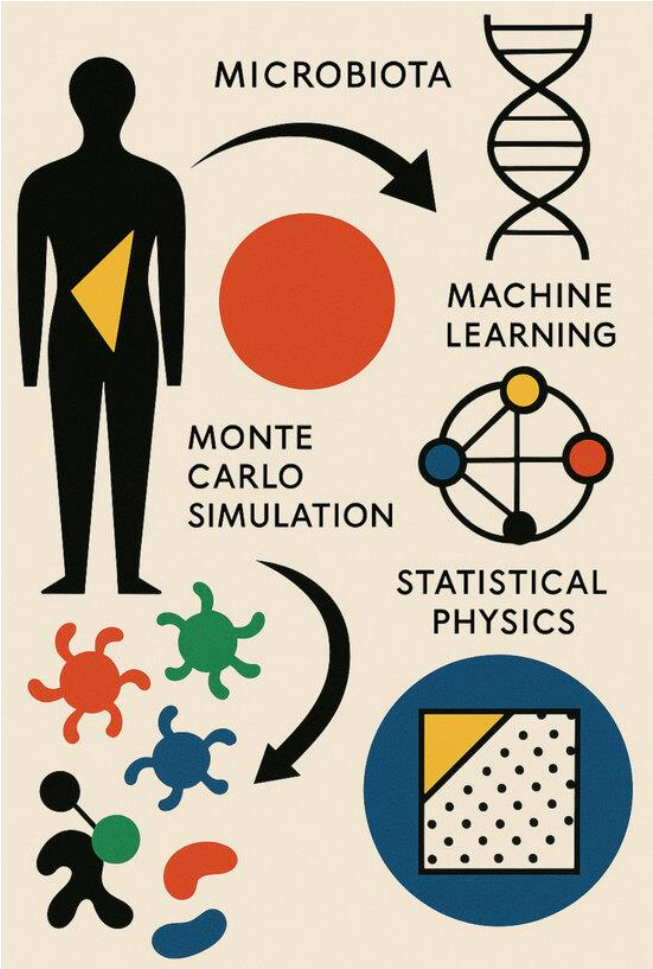
$th < 0.3$

$th > 0.2$
Cirsium arvense



Microbiota

Bellingeri et al. 2024 Mach. Learn. Knowl. Extr.



ISCR
CINECA

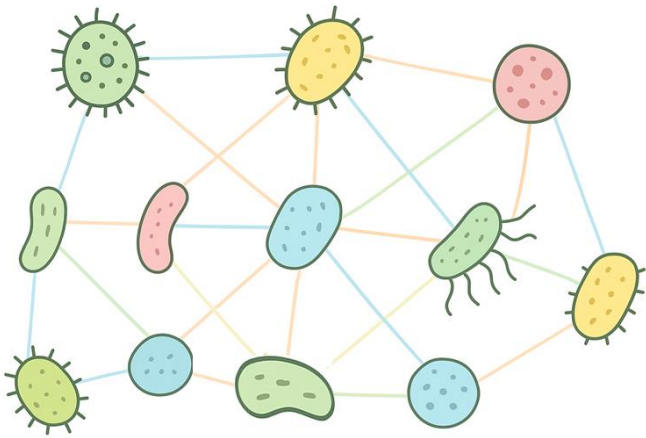


Groups	<i>W</i>	S1	S2	S3	S4	S5	S6
G1	C1	0	0.5	0	0	0	0.5
G1	C2	0.1	0	0.3	0	0	0.6
G1	C3	0	0	0.45	0	0	0.55
G2	C4	0	0	0.4	0	0.6	0
G2	C5	0	0	0.2	0	0.8	0
G3	C6	0	0	0	0.2	0.8	0
G3	C7	0	0	0.2	0.1	0.7	0
G3	C8	0	0	0.05	0.55	0.4	0
G4	C9	0	0.25	0	0.75	0	0
G4	C10	0	1	0	0	0	0

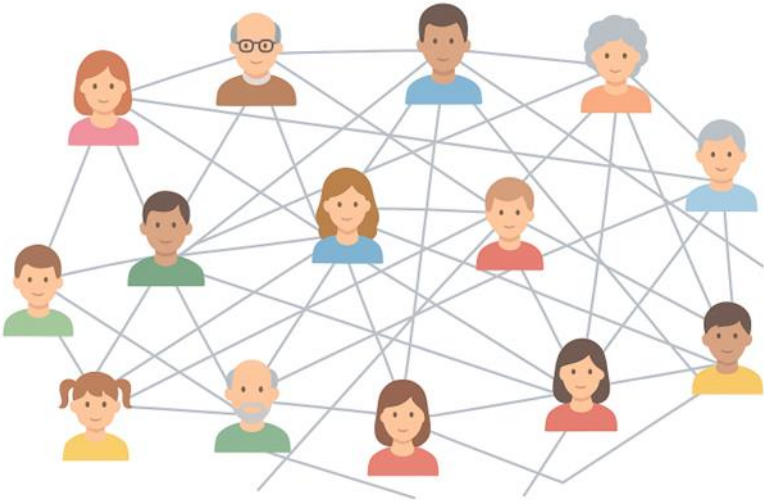
2000 specie

5000 campioni

Network Batteri



Network Pazienti



Il Gruppo



Prof. Davide Cassi



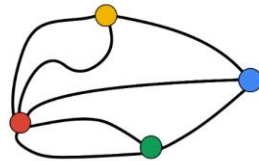
Prof. Roberto Alfieri



Dr. Michele Bellingeri



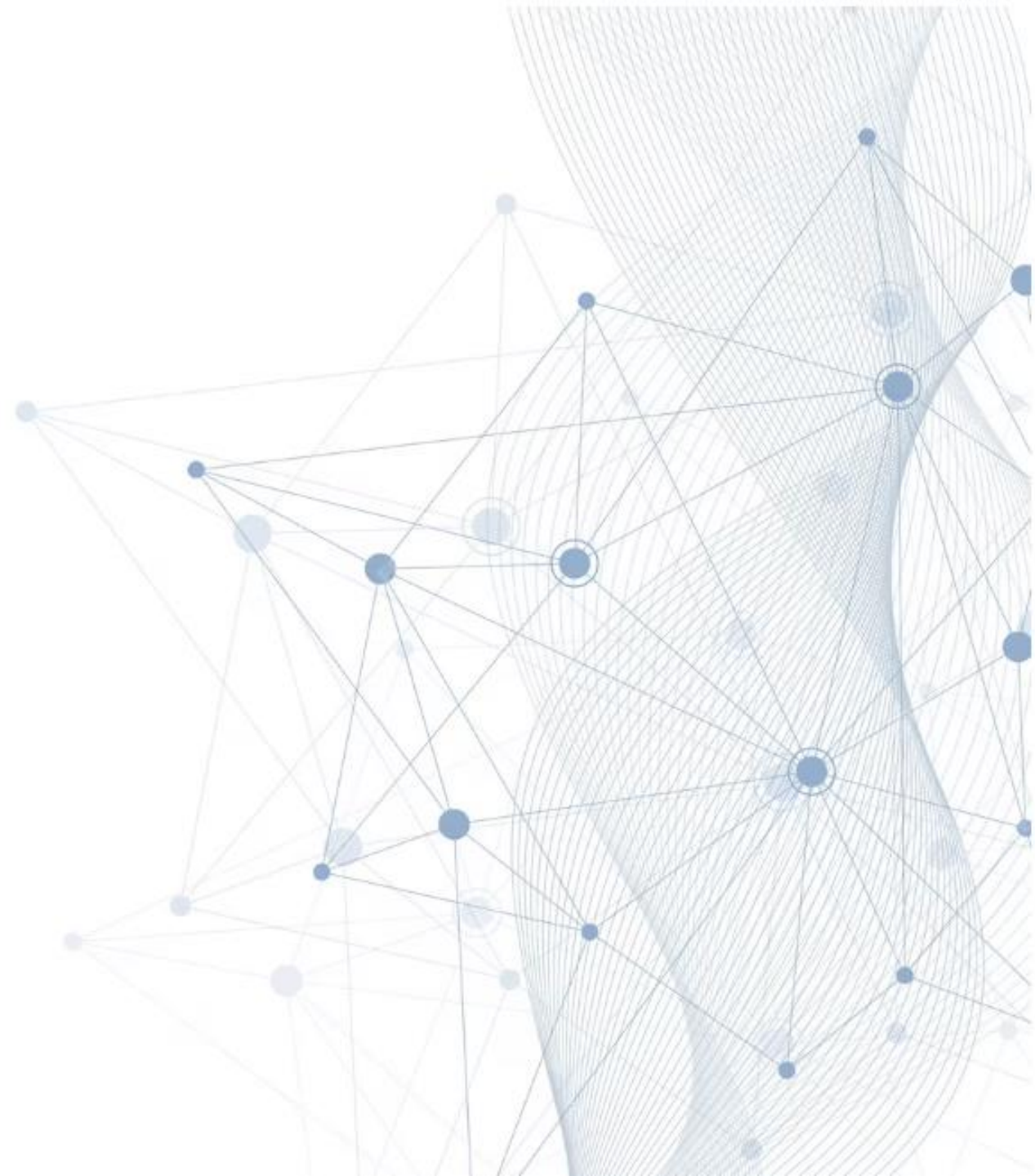
Dr. Massimiliano Turchetto



Grazie

michele.bellingeri@unipr.it

<https://www.networks.unipr.it>



Cucinare rende umani

LUDWIG FEUERBACH

L'UOMO È CIÒ CHE MANGIA

Saggio introduttivo e cura di Andrea Tagliapietra



Bollati Boringhieri

“L’uomo è ciò che mangia”

Unità dell’essere umano sta proprio nell’alimentazione

Unione tra natura e cultura

bios

significava vita concreta dell’individuo,
ma anche generi di sopravvivenza

Natura ↔ **Cultura**



Ludwig Feuerbach (1804 – 1872)

1. Richiamo dei termini della matrice di confusione

Per un problema **binario** (positivo/negativo), la matrice di confusione è:

	Predetto Positivo	Predetto Negativo
Reale Positivo	TP (veri positivi)	FN (falsi negativi)
Reale Negativo	FP (falsi positivi)	TN (veri negativi)

- **TP**: correttamente predetti come positivi
- **FP**: negativi predetti come positivi
- **FN**: positivi predetti come negativi
- **TN**: correttamente predetti come negativi

1. Precisione (Precision)

Indica la **proporzione di predizioni positive corrette** rispetto a tutte le predizioni positive:

$$\text{Precisione} = \frac{\text{Veri Positivi}}{\text{Veri Positivi} + \text{Falsi Positivi}}$$

2. Richiamo (Recall) o Sensibilità

Indica la **proporzione di veri positivi catturati** rispetto a tutti i positivi reali:

$$\text{Richiamo} = \frac{\text{Veri Positivi}}{\text{Veri Positivi} + \text{Falsi Negativi}}$$

3. F1-score

È la **media armonica tra precisione e richiamo**, utile quando vogliamo bilanciare i due valori:

$$F1 = 2 \cdot \frac{\text{Precisione} \cdot \text{Richiamo}}{\text{Precisione} + \text{Richiamo}}$$